

EC601

Control System

Combined Module Notes

Modules 1 – 7

Semester VI • 3L:0T:0P • 3 Credits

CGEC | B.Tech ECE (2023–27)

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Disclaimer / Study Guide Notice:

These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

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Module 1

Introduction to Control Systems

1. Introduction to Control Systems

A **control system** is an interconnection of physical components arranged to *regulate, direct, or command* the behaviour of another system so as to obtain a desired response. The fundamental goal is to **maintain a specified output** in spite of disturbances or parameter variations.

Formal Definition: A control system is a system in which the output quantity is controlled by varying the input quantity. It comprises a *controller*, a *plant* (process), and — in feedback systems — a *sensor* that measures the output.

1.1 Key Terminology

Term	Definition
Plant / Process	The physical object or process to be controlled (e.g. motor, furnace, tank).
Controlled Variable	The output quantity measured and maintained at a desired level (e.g. speed, temperature).
Reference Input $R(s)$	The desired (set-point) value of the controlled variable, supplied externally.
Error Signal $E(s)$	$E = R - B$; difference between reference input and feedback signal. Drives the controller.
Actuator	Device (motor, valve, heater) that physically drives the plant based on controller output.
Sensor / Transducer	Measures the actual output and converts it into a form comparable to the reference signal.
Feedback Signal $B(s)$	Signal derived from the output (via sensor) fed back to the summing junction.
Disturbance	Unwanted external or internal signal that adversely affects the plant output.
Controller	Element that processes the error signal and generates corrective action $U(s)$.

1.2 Industrial Control Examples

1. **Furnace Temperature Control:** Thermostat measures temperature, compares with set-point, switches heater on/off — classic *closed loop on-off* controller.
2. **DC Motor Speed Control:** Tachometer feeds back actual RPM; controller adjusts armature voltage to maintain desired speed despite load variations.
3. **Liquid Level Control:** Float/electronic sensor detects tank level; valve opened/closed to keep level at set value.
4. **Missile Guidance System:** Inertial/radar sensor tracks actual position and continuously corrects trajectory — high-speed closed loop system.
5. **Aircraft Auto-Pilot:** Gyroscopes and accelerometers measure altitude, heading, speed; auto-pilot adjusts control surfaces to match reference values.

2. Transfer Function

The transfer function is the primary mathematical tool to model, analyse, and design LTI control systems. It converts the time-domain differential equation into an algebraic expression in the s -domain (Laplace domain).

Definition: The **Transfer Function** $G(s)$ of a linear time-invariant (LTI) system is the ratio of the Laplace transform of the *output* to the Laplace transform of the *input*, with all initial conditions assumed **zero**.

$$G(s) = \left. \frac{\mathcal{L}\{\text{output}\}}{\mathcal{L}\{\text{input}\}} \right|_{\text{zero I.C.}} = \frac{Y(s)}{U(s)}$$

2.1 Poles, Zeros, and Order

The denominator polynomial set to zero gives the **characteristic equation**. Its roots are the **poles** — they determine stability and transient response. Roots of the numerator are the **zeros**. The **order** is the highest power of s in the denominator.

2.2 Properties of Transfer Function

Properties of Transfer Function	
1	Defined and valid only for LTI (Linear, Time-Invariant) systems.
2	Independent of input magnitude — depends only on system parameters (R, C, L, K , etc.).
3	Setting denominator = 0 gives the characteristic equation ; its roots are the poles .
4	Setting numerator = 0 gives the zeros of the system.
5	Order of system = degree of the denominator polynomial (n).
6	All initial conditions assumed zero — it is an <i>input-output</i> model only.
7	Does not describe internal (state) variables of the system.

2.3 Standard Form

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}, \quad n \geq m \text{ (proper system)}$$

2.4 Example — Series RC Circuit

Given: Series RC circuit. Input = $V_i(s)$, Output = $V_o(s)$ (voltage across C).
Applying voltage divider in s -domain:

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{1/sC}{R + 1/sC} = \frac{1}{1 + sRC} = \frac{1}{1 + s\tau}, \quad \tau = RC$$

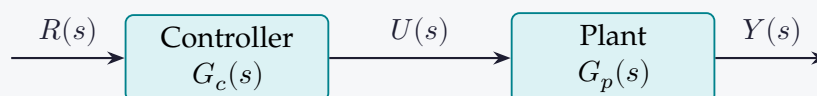
Analysis: First-order system ($n = 1$). One pole at $s = -1/\tau$, no finite zeros. Time constant $\tau = RC$ determines charging speed.

3. Open Loop Control System

In an open loop system, the controller generates a control signal based *only* on the reference input. The actual output is **never measured or compared** — the system cannot self-correct.

Definition: An **open loop control system** is one in which the control action is *independent of the output*. The output has no effect on the control signal.

Open Loop Block Diagram



$$\frac{Y(s)}{R(s)} = G_c(s) \cdot G_p(s)$$

3.1 Characteristics of Open Loop Systems

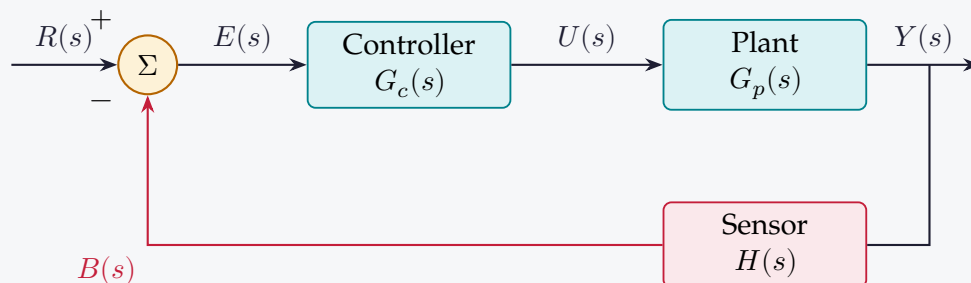
Feature	Open Loop Behaviour
Accuracy	Low — errors from disturbances or parameter changes are not corrected.
Stability	Generally stable; no feedback loop to introduce phase shift or oscillation.
Complexity	Simple design; fewer components; easy to maintain.
Cost	Lower — no sensors, feedback elements, or comparators needed.
Disturbance	Output permanently affected; no self-correction mechanism.
Calibration	Must be re-calibrated frequently to maintain accuracy.
Real Examples	Washing machine (timed cycle), traffic signal timer, electric toaster, stepper motor.

4. Closed Loop (Feedback) Control System

A closed loop system **continuously monitors the output** and uses the measured value to correct the control action. The difference between reference and actual output is the **error**, which drives the controller.

Definition: A closed loop (feedback) control system is one in which the output is measured, compared with the reference input, and the resulting *error signal* is used to reduce the deviation. Feedback is typically **negative**.

Closed Loop Block Diagram (Negative Feedback)



4.1 Closed Loop Transfer Function (CLTF)

Derivation (Unity Feedback, $H(s) = 1$):

$$\begin{aligned} E(s) &= R(s) - Y(s) \\ Y(s) &= G(s) \cdot E(s) = G(s)[R(s) - Y(s)] \\ Y(s)[1 + G(s)] &= G(s) \cdot R(s) \end{aligned}$$

Unity Feedback ($H(s) = 1$):

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

Non-Unity Feedback ($H(s) \neq 1$):

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}, \quad G(s) = G_c(s) \cdot G_p(s)$$

$G(s)H(s)$ = **open loop transfer function (OLTF)**. $1 + G(s)H(s) = 0$ = **characteristic equation**.

4.2 Open Loop vs. Closed Loop — Comparison

Parameter	Open Loop	Closed Loop
Feedback	Absent	Present (negative)
Accuracy	Lower	Higher
Stability	More stable	May become unstable
Disturbance effect	High; no correction	Reduced; self-correcting
Bandwidth	Narrow	Wider
Sensitivity	High to parameter changes	Reduced (robust)
Cost & Complexity	Low	Higher
Real Examples	Toaster, timer, stepper motor	Thermostat, servo drive, auto-pilot

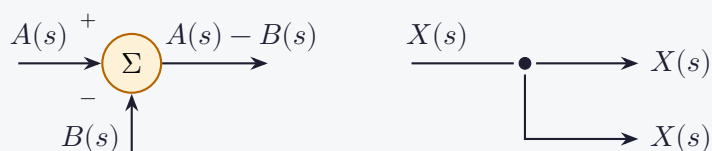
5. Block Diagram Reduction

A **block diagram** is a pictorial representation of a control system. For complex multi-loop systems, **algebraic reduction rules** simplify the diagram step-by-step until a single overall TF $Y(s)/R(s)$ is obtained.

Summing junction: Circle with $+/ -$ signs; algebraically adds/subtracts incoming signals.

Takeoff point: Node from which the same signal is distributed to multiple branches unchanged.

Summing Junction and Takeoff Point



Summing Junction

Takeoff Point

5.1 Reduction Rules

Operation	Description	Equivalent TF
Blocks in Series	Output of G_1 feeds input of G_2 ; same signal chain	$G_1(s) \cdot G_2(s)$
Blocks in Parallel	Same input, individual outputs summed at a junction	$G_1(s) + G_2(s)$
Feedback Loop	Forward path G , feedback path H (negative feedback)	$\frac{G}{1 + GH}$
Summing pt. forward	Shift summing junction past block G to the right	Divide shifted branch by G
Summing pt. backward	Shift summing junction past block G to the left	Multiply shifted branch by G
Takeoff pt. forward	Shift takeoff past block G to the right	Multiply branch by G
Takeoff pt. backward	Shift takeoff past block G to the left	Divide branch by G

5.2 Reduction Procedure (Step-by-Step)

- Step 1:** Identify and combine all **cascade (series)** blocks into a single block.
- Step 2:** Identify and combine all **parallel** paths into a single block.
- Step 3:** Eliminate all **feedback loops** using $G/(1 \pm GH)$.
- Step 4:** If needed, **shift** summing junctions or takeoff points to untangle crossed paths.
- Step 5:** Repeat steps 1–4 until only a **single block** $Y(s)/R(s)$ remains.

Exam Tip

Always maintain the correct sign (+/−) at every summing junction while shifting. Careless sign errors are the most common mistake in block diagram problems.

6. Signal Flow Graph (SFG) Analysis

A Signal Flow Graph is an alternative graphical method to represent and solve linear algebraic equations describing a control system. Unlike block diagrams — which require step-by-step reduction — an SFG is solved *directly* using **Mason's Gain Formula** in a single pass.

Definition: A **Signal Flow Graph (SFG)** is a directed graph of *nodes* (signals/variables) connected by *directed branches* (carrying gain/transmittance values). It is a compact representation of the system's simultaneous equations.

6.1 Basic Terminology

Term	Meaning / Definition
Node	Represents a signal or variable in the system (shown as a small circle).
Branch	Directed arrow between two nodes; labelled with the <i>transmittance</i> (gain).
Source Node	Input node — has only outgoing branches; no incoming branches.
Sink Node	Output node — has only incoming branches; no outgoing branches.
Mixed Node	A node with both incoming and outgoing branches (intermediate variable).
Forward Path	Continuous directed path from source to sink in which no node is visited more than once .
Forward Path Gain P_k	Product of all branch gains along the k -th forward path.
Loop	Closed directed path beginning and ending at the same node, no node repeated .
Loop Gain L_i	Product of all branch gains around the i -th loop.
Non-Touching Loops	Two or more loops that do not share any common node .

6.2 Mason's Gain Formula

$$T = \frac{Y(s)}{R(s)} = \frac{\sum_{k=1}^N P_k \Delta_k}{\Delta}$$

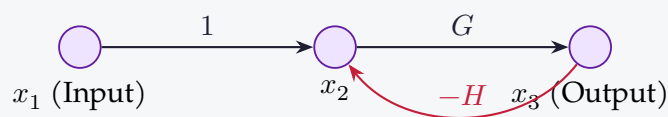
Symbol	Definition
N	Total number of forward paths from source to sink.
P_k	Gain of the k -th forward path (product of branch gains along that path).
Δ	Graph determinant = $1 - \sum_i L_i + \sum_{i,j} L_i L_j - \sum_{i,j,k} L_i L_j L_k + \dots$
$\sum L_i$	Sum of gains of all individual loops in the graph.
$\sum L_i L_j$	Sum of gain-products of all pairs of non-touching loops .
$\sum L_i L_j L_k$	Sum of gain-products of all triplets of mutually non-touching loops .
Δ_k	Cofactor of the k -th path: value of Δ after removing all loops that <i>touch</i> forward path k .

6.3 Steps to Apply Mason's Formula

- 1. Identify forward paths:** Trace every path from source to sink (no repeated nodes). Note gain P_k of each.
- 2. Identify all loops:** Find every closed path. Compute loop gain L_i for each.
- 3. Find non-touching loop combinations:** Identify all pairs, triplets, etc. sharing no common node.
- 4. Compute graph determinant:** $\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k + \dots$
- 5. Compute cofactor Δ_k :** For each forward path k , remove all loops touching that path from Δ .
- 6. Apply Mason's formula:** $T = \frac{\sum P_k \Delta_k}{\Delta}$.

6.4 Worked Example — Simple Feedback System

SFG Representation



Applying Mason's formula:

- Forward path: $P_1 = 1 \times G = G$
- Loop: $L_1 = G \times (-H) = -GH$
- No non-touching loop pairs (only one loop).
- $\Delta = 1 - L_1 = 1 - (-GH) = 1 + GH$
- $\Delta_1 = 1$ (loop L_1 touches forward path P_1 , so removed from Δ)

$$T = \frac{P_1 \Delta_1}{\Delta} = \frac{G \cdot 1}{1 + GH} = \frac{G}{1 + GH} \quad \checkmark$$

Matches the CLTF from block diagram reduction — both methods are equivalent.

6.5 SFG vs. Block Diagram — Comparison

Aspect	Block Diagram	Signal Flow Graph
Representation	Blocks, summing junctions, takeoff points	Directed nodes and weighted branches
Reduction Method	Step-by-step algebra; multiple passes	Mason's formula — single direct computation
Complex Systems	Tedious; can become cluttered and error-prone	Compact, systematic, less error-prone
Gain Formula	No direct formula; reduce iteratively	$T = \sum P_k \Delta_k / \Delta$
Preferred For	Simple systems; visual understanding	Multi-loop systems; quick TF computation

Quick Revision — Module 1

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Key Formulas at a Glance

Transfer Function	$G(s) = Y(s)/U(s)$ (zero initial conditions)
CLTF (Unity feedback)	$T(s) = G(s)/[1 + G(s)]$
CLTF (Non-unity)	$T(s) = G(s)/[1 + G(s)H(s)]$
Characteristic Equation	$1 + G(s)H(s) = 0$
Series blocks	$G_{eq} = G_1 \cdot G_2$
Parallel blocks	$G_{eq} = G_1 + G_2$
Feedback reduction	$G_{eq} = G/(1 \pm GH)$
Mason's Gain Formula	$T = \frac{\sum_k P_k \Delta_k}{\Delta}$
Graph Determinant	$\Delta = 1 - \sum L_i + \sum L_i L_j - \dots$

One-Line Definitions

- **Control System:** Interconnection of components to regulate output to a desired value.
- **Transfer Function:** Ratio of Laplace output to Laplace input at zero initial conditions.
- **Poles:** Roots of denominator of $G(s)$ — determine stability and transient response.
- **Zeros:** Roots of numerator of $G(s)$ — affect frequency response shape.
- **Open Loop:** Control action independent of output; no self-correction.
- **Closed Loop:** Output fed back and compared with reference; error drives controller.
- **Error Signal:** $E(s) = R(s) - B(s)$; difference between reference and feedback.
- **SFG:** Directed graph of nodes and branches representing system equations.
- **Forward Path:** Path from source to sink with no repeated nodes.
- **Loop Gain:** Product of all branch gains around a closed loop.
- **Cofactor Δ_k :** Graph determinant Δ with all loops touching path k removed.

Frequently Asked / Viva Points

Q1. Why are initial conditions zero in TF?

TF is an input-output model; it characterises the system, not a specific response.

Frequently Asked / Viva Questions (contd.)

Zero I.C. ensures the TF is a property of the system alone.

Q2. What is the order of a system?

Highest power of s in the denominator of $G(s)$.

Q3. Difference between open loop and closed loop?

Open loop: no feedback, no self-correction. Closed loop: feedback present, error-driven, self-correcting.

Q4. What is the characteristic equation?

$1 + G(s)H(s) = 0$. Its roots (poles of CLTF) determine stability.

Q5. When is a system proper?

When degree of denominator $\geq$ degree of numerator ($n \geq m$).

Q6. What is a takeoff point?

A node from which the same signal branches to multiple paths without change.

Q7. What is Δ_k in Mason's formula?

Cofactor of the k -th forward path — Δ computed after removing all loops that touch path k .

Q8. Can SFG and block diagram give different results?

No — both are equivalent representations; Mason's formula gives the same TF as block diagram reduction.

Q9. What are non-touching loops?

Loops that share no common node. Their gain products appear in Δ .

Q10. Why is negative feedback preferred?

It reduces error, improves accuracy, reduces sensitivity to parameter variations, and widens bandwidth.

Comparison Snapshot

Property	Open Loop	Closed Loop
Feedback	No	Yes (negative)
Accuracy	Low	High
Stability	More stable	Can oscillate
Cost	Low	High
Disturbance	Not corrected	Self-corrected

Module 2

Stability of Feedback Control Systems

1. Stability of Feedback Control Systems

Stability is the most fundamental requirement of any control system. An unstable system produces unbounded output even for a bounded input, making it practically useless.

BIBO Stability: A system is **Bounded-Input Bounded-Output (BIBO) stable** if every bounded input produces a bounded output. For an LTI system, this requires all poles of the closed-loop transfer function to lie in the **left half of the s -plane** (i.e., have negative real parts).

1.1 Stability Conditions

Condition	Meaning
Stable	All closed-loop poles have negative real parts (left half s -plane). Output decays to zero.
Marginally Stable	Poles on the imaginary axis (zero real part). Output oscillates with constant amplitude.
Unstable	At least one pole in the right half s-plane . Output grows without bound.

1.2 Relative Stability

Absolute stability only tells us *whether* a system is stable. **Relative stability** tells us *how stable* — i.e., how far the closed-loop poles are from the imaginary axis.

Relative Stability: A measure of the degree of stability. A system with poles far into the left half-plane is *more stable* (faster decay, less oscillation) than one with poles close to the imaginary axis. Quantified by **gain margin** and **phase margin** (covered in Module 3).

Key Concept

If all poles of $T(s) = \frac{G(s)}{1 + G(s)H(s)}$ satisfy $\text{Re}(s_i) < 0$, the system is stable.

The **characteristic equation** $1 + G(s)H(s) = 0$ determines the pole locations.

2. Routh–Hurwitz Stability Criterion

The Routh–Hurwitz criterion determines stability **without solving** the characteristic equation. It counts the number of roots in the right half s -plane using a tabular method.

Definition: The **Routh–Hurwitz criterion** states that a system is stable if and only if all elements of the **first column** of the Routh array are **positive**. The number of sign changes in the first column equals the number of closed-loop poles in the right half s -plane.

2.1 Constructing the Routh Array

Given the characteristic equation:

$$a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0$$

Step 1 — Necessary condition: All coefficients a_i must be present and have the **same sign**. If any coefficient is zero or negative, the system is *unstable or marginally stable* (no need to proceed).

Step 2 — Form the array:

Routh Array Structure

s^n	a_n	a_{n-2}	a_{n-4}	$\dots$
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	$\dots$
s^{n-2}	b_1	b_2	b_3	$\dots$
s^{n-3}	c_1	c_2	c_3	$\dots$
$\vdots$	$\vdots$	$\vdots$		
s^0	a_0			

Row Computation Formulas

$$b_1 = \frac{a_{n-1} \cdot a_{n-2} - a_n \cdot a_{n-3}}{a_{n-1}}, \quad b_2 = \frac{a_{n-1} \cdot a_{n-4} - a_n \cdot a_{n-5}}{a_{n-1}}, \quad \dots$$

$$c_1 = \frac{b_1 \cdot a_{n-3} - a_{n-1} \cdot b_2}{b_1}, \quad c_2 = \frac{b_1 \cdot a_{n-5} - a_{n-1} \cdot b_3}{b_1}, \quad \dots$$

Continue until all rows are complete. **Stability:** No sign changes in column 1.

2.2 Special Cases

Special Case	Handling
Zero in first column	Replace zero with small $\varepsilon > 0$, complete the array, then let $\varepsilon \rightarrow 0^+$. Count sign changes.
Entire row of zeros	System has symmetric roots (pairs on imaginary axis or real axis). Form auxiliary polynomial from the row above, differentiate it, and use its coefficients to replace the zero row.

2.3 Worked Example

Example — 3rd Order System

Characteristic equation: $s^3 + 2s^2 + 4s + 8 = 0$

Routh Array:

$$\begin{array}{c|cc}
 s^3 & 1 & 4 \\
 s^2 & 2 & 8 \\
 s^1 & b_1 = \frac{2 \times 4 - 1 \times 8}{2} = \frac{0}{2} = 0 & 0 \\
 s^0 & 8 &
 \end{array}$$

$b_1 = 0 \rightarrow$ zero in first column $\rightarrow$ replace with ε , let $\varepsilon \rightarrow 0^+$.

First column: 1, 2, ε , 8 — as $\varepsilon \rightarrow 0^+$, no sign change.

Result: Marginally stable (one pair of roots on imaginary axis at $s = \pm j2$).

Exam Tips — Routh Criterion

- **Necessary condition first:** All coefficients positive and non-zero — check before building the array.
- **Number of sign changes** in column 1 = number of unstable poles.
- **Marginal stability:** Entire row of zeros $\rightarrow$ auxiliary polynomial roots on $j\omega$ -axis.
- **Range of K for stability:** Set up array with K , find condition on first column > 0 .
- Routh criterion gives *no information* about the actual pole locations — only their count in RHP.

3. Steady-State Error (SSE)

Even a stable system may not track the reference input perfectly in steady state. The **steady-state error** quantifies this residual deviation after all transients have died out.

Definition: The **steady-state error** e_{ss} is the difference between the desired reference input and the actual output as $t \rightarrow \infty$:

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s)$$

where $E(s) = \frac{R(s)}{1 + G(s)H(s)}$ for a unity-feedback system.

3.1 System Type Number

The **type number** of a system is the number of open-loop poles at the origin (i.e., the number of pure integrators in $G(s)H(s)$).

$$G(s)H(s) = \frac{K(1 + T_1s)(1 + T_2s) \cdots}{s^N(1 + T_as)(1 + T_bs) \cdots}$$

Here N is the **type number** (0, 1, 2, ...). More integrators $\rightarrow$ better steady-state tracking.

3.2 Error Constants and SSE Formulas

Input	Type 0	Type 1	Type 2
Step $R(s) = \frac{1}{s}$	$\frac{1}{1 + K_p}$	0	0
Ramp $R(s) = \frac{1}{s^2}$	∞	$\frac{1}{K_v}$	0
Parabola $R(s) = \frac{1}{s^3}$	∞	∞	$\frac{1}{K_a}$

Error Constants

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) \quad (\text{Position constant})$$

$$K_v = \lim_{s \rightarrow 0} s G(s)H(s) \quad (\text{Velocity constant})$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) \quad (\text{Acceleration constant})$$

3.3 Steady-State Accuracy

Steady-State Accuracy is the ability of a control system to track the reference input with minimum error in steady state. It is directly related to the **type number** and the **gain** of the open-loop transfer function. Higher type number and higher gain both improve steady-state accuracy.

Exam Tips — SSE

- SSE depends on **system type** and **input type** — memorise the table above.
- Increasing open-loop gain K reduces SSE for the same type, but may reduce stability.
- Adding an integrator (increasing type) eliminates SSE for lower-order inputs but risks instability.
- For **non-unity feedback**, use $E(s) = R(s) - C(s)$ and apply final value theorem carefully.
- **Final Value Theorem** valid only if $sE(s)$ has all poles in LHP (system must be stable).

4. Disturbance Rejection, Sensitivity and Robustness

4.1 Disturbance Rejection

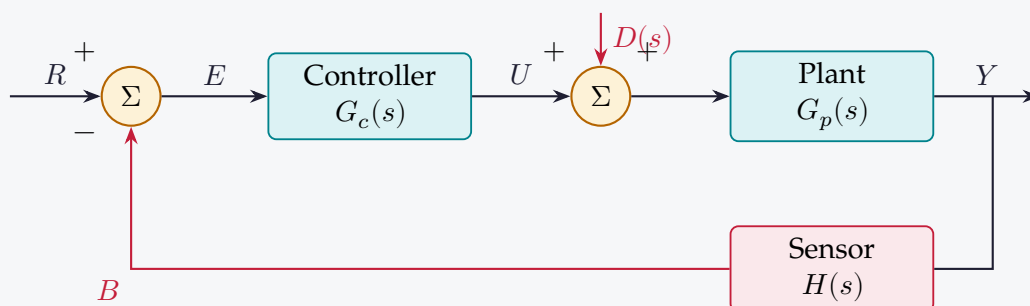
Real systems are subject to unwanted inputs (disturbances) such as load changes, noise, and environmental variations. A good control system must **reject disturbances** — i.e., minimise their effect on the output.

Disturbance Rejection: The ability of a closed-loop system to suppress the effect of disturbance inputs $D(s)$ on the output $Y(s)$. The **disturbance transfer function** is:

$$\left. \frac{Y(s)}{D(s)} \right|_{R=0} = \frac{G_p(s)}{1 + G_c(s)G_p(s)H(s)}$$

High controller gain $G_c(s)$ reduces this ratio, improving disturbance rejection.

Closed Loop with Disturbance



4.2 Sensitivity

Sensitivity Function $S(s)$: Measures how much the closed-loop transfer function $T(s)$ changes relative to changes in the plant $G(s)$:

$$S(s) = \frac{1}{1 + G(s)H(s)}$$

Complementary Sensitivity $T(s) = 1 - S(s) = \frac{G(s)H(s)}{1 + G(s)H(s)}$

Note: $S(s) + T(s) = 1$ always. High loop gain $\Rightarrow S \approx 0$ (low sensitivity).

4.3 Robustness

Robustness: The ability of a control system to maintain acceptable performance despite *uncertainties* in the plant model (parameter variations, unmodelled dynamics, external disturbances). A robust system has:

- Low sensitivity to plant parameter changes
- Good disturbance rejection
- Adequate gain and phase margins

Property	Open Loop	Closed Loop
Sensitivity to ΔG	High — directly affects output	Reduced by factor $1 + GH$
Disturbance rejection	None	Good (high gain helps)
Robustness	Poor	Better (with proper design)
Bandwidth	Narrow	Wider

5. PID Controllers

A **PID controller** combines three control actions — Proportional, Integral, and Derivative — to achieve fast response, zero steady-state error, and good damping simultaneously.

PID Control Law (Time Domain)

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Transfer Function (s-domain):

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s}$$

5.1 Proportional (P) Controller

P Controller: $u(t) = K_p e(t)$, $G_c(s) = K_p$

Output is proportional to the current error. Increasing K_p reduces SSE but increases overshoot and may cause instability.

Effect of increasing K_p	Result
Rise time	Decreases
Overshoot	Increases
Settling time	Small change
Steady-state error	Decreases (but never zero for Type 0)
Stability	Degrades at very high K_p

5.2 Integral (I) Controller

I Controller: $u(t) = K_i \int e(t) dt$, $G_c(s) = K_i/s$

Integrates the error over time. Eliminates steady-state error completely (adds a pole at origin, increasing system type by 1). However, it slows the response and can cause instability (integral windup).

5.3 Derivative (D) Controller

D Controller: $u(t) = K_d \frac{de}{dt}$, $G_c(s) = K_d s$

Responds to the *rate of change* of error — anticipates future error. Improves damping and reduces overshoot. **Never used alone** (amplifies noise; no steady-state correction). Always combined with P or PI.

5.4 Combined PID — Effect Summary

Controller	Rise Time	Overshoot	Settling	SSE
P	Decrease	Increase	Small Δ	Decrease
I	Decrease	Increase	Increase	Eliminate
D	Small Δ	Decrease	Decrease	No effect
PI	Decrease	Increase	Increase	Eliminate
PD	Decrease	Decrease	Decrease	No effect
PID	Decrease	Decrease	Decrease	Eliminate

Exam Tips — PID

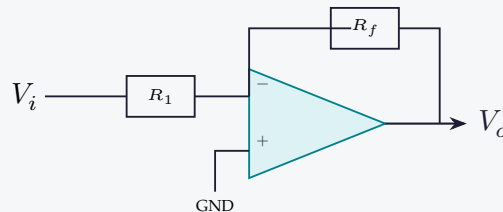
- **P alone:** Always has residual SSE for Type 0 systems.
- **I action:** Eliminates SSE but causes *integral windup* — output saturates if error persists.
- **D action:** Improves transient response; **never used alone** — amplifies high-frequency noise.
- **PID transfer function** has a zero at origin (from K_i/s) and two zeros from numerator.
- **Ziegler–Nichols method** is the standard tuning technique for PID (may appear in viva).

6. Realization of PID Controllers

6.1 Op-Amp Realization

Op-amp circuits implement P, I, D, and PID actions using resistors and capacitors. The inverting op-amp configuration is standard.

Proportional Controller

P Controller — Inverting Amplifier

$$G_c(s) = -\frac{R_f}{R_1} = -K_p \quad (K_p = R_f/R_1)$$

PI Controller

Replace R_f with a series combination of R_f and capacitor C_f :

PI Controller — Op-Amp Transfer Function

$$G_c(s) = -\frac{Z_f}{Z_1} = -\frac{R_f + 1/(sC_f)}{R_1} = -\left(\frac{R_f}{R_1} + \frac{1}{sR_1C_f}\right) = -\left(K_p + \frac{K_i}{s}\right)$$

$$K_p = \frac{R_f}{R_1}, \quad K_i = \frac{1}{R_1C_f}$$

PD Controller

Replace input R_1 with a series combination of R_1 and capacitor C_1 :

PD Controller — Op-Amp Transfer Function

$$G_c(s) = -\frac{R_f}{R_1 + 1/(sC_1)} = -\frac{sR_fC_1}{1 + sR_1C_1} \approx -(K_p + K_d s) \text{ for small } R_1$$

$$K_p = \frac{R_f}{R_1}, \quad K_d = R_fC_1$$

PID Controller (Op-Amp)

PID Controller — Op-Amp Transfer Function

Use $Z_1 = R_1 + 1/(sC_1)$ as input impedance and $Z_f = R_f \parallel (1/sC_f)$ as feedback impedance:

$$G_c(s) = -\frac{Z_f}{Z_1} = -\frac{R_f/(1 + sR_fC_f)}{R_1 + 1/(sC_1)}$$

Simplified ideal form:

$$G_c(s) = -\left(K_p + \frac{K_i}{s} + K_d s\right)$$

$$K_p = \frac{R_f}{R_1}, \quad K_i = \frac{1}{R_1C_f}, \quad K_d = R_fC_1$$

The negative sign is corrected by cascading an inverting unity-gain stage.

6.2 Digital Implementation of PID

In digital (discrete-time) systems, the PID algorithm is implemented in software using numerical approximations of integration and differentiation.

Discrete PID Algorithm: With sampling period T_s , the continuous PID is discretised as:

- **Proportional:** $P[k] = K_p e[k]$
- **Integral** (rectangular/trapezoidal): $I[k] = I[k-1] + K_i T_s e[k]$
- **Derivative** (backward difference): $D[k] = K_d \frac{e[k] - e[k-1]}{T_s}$
- **Output:** $u[k] = P[k] + I[k] + D[k]$

Pulse Transfer Function (Z-domain)

Using backward Euler ($s \approx \frac{z-1}{T_s z}$):

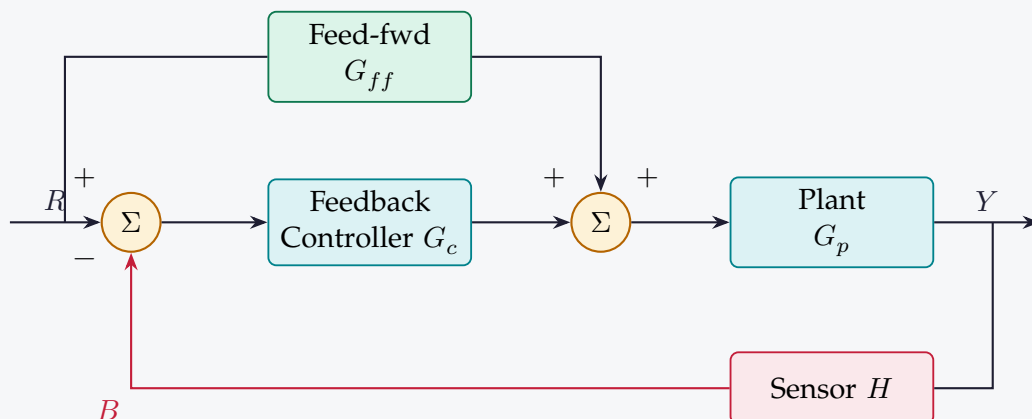
$$G_c(z) = K_p + K_i \frac{T_s z}{z-1} + K_d \frac{z-1}{T_s z}$$

Aspect	Analog (Op-Amp)	Digital (Microcontroller)
Implementation	Hardware (R, C, op-amp)	Software (algorithm in MCU/DSP)
Tuning	Change component values	Change K_p , K_i , K_d in code
Noise	Susceptible to component drift	Quantisation noise; sampling delay
Flexibility	Fixed once built	Easily reprogrammable
Cost	Low for simple systems	Higher (processor needed)

7. Feed-Forward Control

Feed-Forward Control: A control strategy in which the controller acts on a *measured disturbance* or the *reference input directly*, before the disturbance affects the output. It does **not** use feedback — it anticipates and cancels the disturbance proactively.

Feed-Forward + Feedback Configuration



Ideal Feed-Forward Condition

For perfect disturbance cancellation, the feed-forward controller should satisfy:

$$G_{ff}(s) = -\frac{G_d(s)}{G_p(s)}$$

where $G_d(s)$ is the disturbance transfer function and $G_p(s)$ is the plant. In practice, exact cancellation is impossible due to model uncertainty — hence feedback is always retained alongside feed-forward.

Aspect	Feedback Only	Feed-Forward + Feedback
Disturbance handling	Reacts after output is affected	Anticipates and cancels disturbance
Model dependency	Low (error-driven)	High (needs accurate plant model)
Stability risk	Inherent (loop gain)	Lower (FF is open-loop)
Typical use	General purpose	Known, measurable disturbances

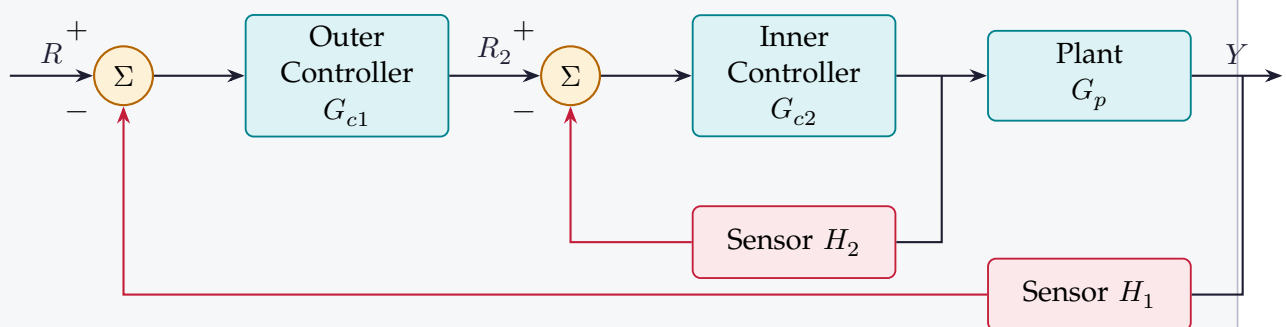
8. Multi-Loop Control Configurations

Multi-Loop Control: Control architectures that use more than one feedback loop to improve performance. The most common configurations are **cascade (series)** control and **ratio control**.

8.1 Cascade (Inner–Outer Loop) Control

Cascade Control: Two feedback loops — an **inner (secondary) loop** and an **outer (primary) loop**. The output of the outer controller becomes the *set-point* for the inner controller. The inner loop handles fast disturbances; the outer loop handles the primary variable.

Cascade Control Block Diagram



Advantages of Cascade Control:

- Inner loop rejects disturbances *before* they propagate to the outer loop.
- Faster overall response — inner loop is tuned for speed.
- Outer loop can be tuned independently for accuracy.

8.2 Ratio Control

Ratio Control: Maintains a fixed ratio between two process variables (e.g., fuel-to-air ratio in a combustion system). One variable is the *wild* (uncontrolled) stream; the other is the *controlled* stream adjusted to maintain the ratio.

$$\text{Set-point of controlled stream} = K_r \times (\text{measured wild stream})$$

Quick Revision — Module 2

Feedback Control Systems | EC601

Key Formulas at a Glance

BIBO Stability	All closed-loop poles in LHP ($\text{Re}(s) < 0$)
Characteristic Equation	$1 + G(s)H(s) = 0$
Sensitivity Function	$S(s) = \frac{1}{1 + G(s)H(s)}$
Disturbance TF	$\frac{Y}{D}\bigg _{R=0} = \frac{G_p}{1 + G_c G_p H}$
SSE (step, Type 0)	$e_{ss} = \frac{1}{1 + K_p}$
SSE (ramp, Type 1)	$e_{ss} = \frac{1}{K_v}$
Position constant	$K_p = \lim_{s \rightarrow 0} G(s)H(s)$
Velocity constant	$K_v = \lim_{s \rightarrow 0} sG(s)H(s)$
PID (s-domain)	$G_c(s) = K_p + K_i/s + K_d s$
PI op-amp	$K_p = R_f/R_1, \quad K_i = 1/(R_1 C_f)$
PID op-amp	$K_p = R_f/R_1, \quad K_i = 1/(R_1 C_f), \quad K_d = R_f C_1$
Digital PID integral	$I[k] = I[k-1] + K_i T_s e[k]$
Digital PID derivative	$D[k] = K_d(e[k] - e[k-1])/T_s$
Ideal FF condition	$G_{ff} = -G_d/G_p$

One-Line Definitions

- **BIBO Stability:** Bounded input $\Rightarrow$ bounded output; all CLTF poles in LHP.
- **Relative Stability:** How far poles are from imaginary axis; measured by gain/phase margin.
- **Routh Criterion:** Algebraic test for stability — no sign changes in first column of Routh array.
- **SSE:** Residual error as $t \rightarrow \infty$; depends on system type and input type.
- **System Type:** Number of open-loop poles at origin (integrators in $G(s)H(s)$).
- **Sensitivity:** $S = 1/(1 + GH)$; measures effect of plant variation on closed-loop TF.
- **Robustness:** Ability to maintain performance despite model uncertainty and disturbances.
- **P Controller:** Proportional to error; reduces SSE but cannot eliminate it (Type 0).

- **I Controller:** Integrates error; eliminates SSE; risks integral windup and instability.
- **D Controller:** Responds to rate of change; improves damping; never used alone.
- **PID:** Combines all three — fast response, zero SSE, good damping.
- **Feed-Forward:** Acts on disturbance before it affects output; requires accurate plant model.
- **Cascade Control:** Inner loop handles fast disturbances; outer loop controls primary variable.

Frequently Asked / Viva Questions

Q1. What is the necessary condition for stability using Routh?

All coefficients of the characteristic polynomial must be positive and non-zero.

Q2. What does a sign change in Routh's first column indicate?

Each sign change corresponds to one closed-loop pole in the right half s -plane.

Q3. What is the difference between absolute and relative stability?

Absolute: stable or not. Relative: how stable (distance of poles from imaginary axis).

Q4. Why does adding an integrator eliminate SSE?

It increases system type by 1, making the velocity/position constant infinite for the corresponding input.

Q5. Why is D action never used alone?

It has no DC gain ($G_c(0) = 0$), cannot correct steady-state error, and amplifies high-frequency noise.

Q6. What is integral windup?

When the integrator accumulates a large value during saturation, causing a large overshoot when the system comes out of saturation.

Q7. What is the sensitivity function?

$S(s) = 1/(1 + GH)$; measures how closed-loop TF changes with plant variation. $S + T = 1$.

Q8. What is the advantage of cascade control?

Inner loop rejects disturbances faster before they affect the outer (primary) loop.

Q9. What is the ideal feed-forward condition?

$G_{ff} = -G_d/G_p$ — cancels disturbance perfectly (requires exact plant model).

Q10. How is PID implemented digitally?

Using backward Euler or Tustin (bilinear) approximation; integral is a running sum, derivative is a backward difference.

SSE Quick Reference Table

Input	Type 0	Type 1	Type 2
Step	$\frac{1}{1 + K_p}$	0	0
Ramp	∞	$\frac{1}{K_v}$	0
Parabola	∞	∞	$\frac{1}{K_a}$

Module 3

Time Response of Second-Order Systems

1. Time Response of Second-Order Systems

A **second-order system** is the most important standard form in control engineering. Its response to a unit step input exhibits all the key transient characteristics — rise, overshoot, oscillation, and settling — that are used to specify and evaluate system performance.

Standard Second-Order Transfer Function:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

where ω_n = **natural frequency** (rad/s) and ζ = **damping ratio** (dimensionless).
The **characteristic equation** is $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$.

1.1 Poles of the Second-Order System

The two closed-loop poles are:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

For $0 < \zeta < 1$ (underdamped): $s_{1,2} = -\zeta\omega_n \pm j\omega_d$, where $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ is the **damped natural frequency**.

1.2 Damping Cases

Condition	Pole Location	Response
$\zeta = 0$	$\pm j\omega_n$ (imaginary)	Undamped — sustained sinusoidal oscillation
$0 < \zeta < 1$	Complex conjugate in LHP	Underdamped — oscillatory, decaying; <i>most common in practice</i>
$\zeta = 1$	Two equal real poles	Critically damped — fastest non-oscillatory response
$\zeta > 1$	Two distinct real poles	Overdamped — sluggish, no oscillation

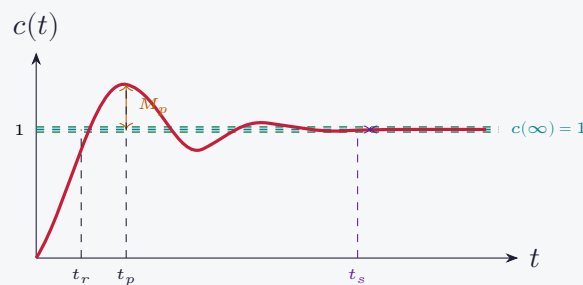
1.3 Unit Step Response (Underdamped, $\zeta < 1$)

Step Response Formula

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi), \quad t \geq 0$$

where $\phi = \cos^{-1}(\zeta)$ and $\omega_d = \omega_n \sqrt{1-\zeta^2}$.

Typical Underdamped Step Response



2. Performance Specifications in Time Domain

Time-domain specifications quantify the transient and steady-state behaviour of a system's step response. They are used to set design requirements and evaluate controller performance.

2.1 Transient Response Specifications

Specification	Symbol	Definition & Formula
Rise Time	t_r	Time for response to rise from 10% to 90% of final value (underdamped); or 0% to 100% (overdamped). $t_r \approx \frac{1.8}{\omega_n}$
Peak Time	t_p	Time to reach the first (maximum) peak of overshoot. $t_p = \frac{\pi}{\omega_d}$
Peak Overshoot	M_p	Maximum overshoot above final value, expressed as %. $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$
Settling Time	t_s	Time for response to stay within $\pm 2\%$ (or $\pm 5\%$) of final value. $t_s \approx \frac{4}{\zeta\omega_n}$ (2% criterion)
Delay Time	t_d	Time for response to reach 50% of final value for the first time. $t_d \approx \frac{1 + 0.7\zeta}{\omega_n}$

Key Relations — Underdamped 2nd Order

$\omega_d = \omega_n \sqrt{1 - \zeta^2}$	Damped natural frequency
$t_p = \pi / \omega_d$	Peak time
$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$	% Peak overshoot
$t_s \approx 4 / (\zeta\omega_n)$	Settling time (2% band)
$t_s \approx 3 / (\zeta\omega_n)$	Settling time (5% band)
$\zeta = \cos \theta$	θ = angle of pole from $-\sigma$ axis

Exam Tips — Time Domain Specs

- **Higher** $\omega_n \rightarrow$ faster response (smaller t_r, t_p, t_s).
- **Higher** $\zeta \rightarrow$ less overshoot, more damping, slower rise.
- $\zeta = 0.707$ gives $\approx 4.3\%$ overshoot — often used as optimal damping.
- M_p **depends only on** ζ ; t_p depends on ω_d ; t_s depends on $\zeta\omega_n$.
- For a 2nd order system, poles at $s = -\zeta\omega_n \pm j\omega_d$ — draw on s -plane for viva.

3. Steady-State Error and Error Constants

Steady-State Error: The difference between the desired reference and the actual output as $t \rightarrow \infty$:

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)H(s)}$$

Valid only when the closed-loop system is **stable** (Final Value Theorem condition).

3.1 Error Constants

Static Error Constants

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) \quad (\text{Position — for step input})$$

$$K_v = \lim_{s \rightarrow 0} s G(s)H(s) \quad (\text{Velocity — for ramp input})$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) \quad (\text{Acceleration — for parabolic input})$$

3.2 SSE vs System Type

Input	Type 0	Type 1	Type 2
Step $\left(\frac{1}{s}\right)$	$\frac{1}{1 + K_p}$	0	0
Ramp $\left(\frac{1}{s^2}\right)$	∞	$\frac{1}{K_v}$	0
Parabola $\left(\frac{1}{s^3}\right)$	∞	∞	$\frac{1}{K_a}$

Exam Tips — SSE

- **System type** = number of open-loop poles at $s = 0$ (pure integrators).
- Increasing gain K reduces SSE but may reduce stability margin.
- Adding an integrator (increasing type) eliminates SSE for lower-order inputs.
- **FVT applies only if** $sE(s)$ has all poles in LHP — check stability first.
- For non-unity feedback: $e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)H(s)}$.

4. Root Locus Method

The **Root Locus** is a graphical method that shows how the closed-loop poles move in the s -plane as the open-loop gain K varies from 0 to ∞ . It is the most powerful tool for **controller**

design and stability analysis.

Definition: The root locus is the locus (path) of the closed-loop poles of the system $T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$ as K varies from 0 to $+\infty$.

Characteristic equation: $1 + KG(s)H(s) = 0 \Rightarrow G(s)H(s) = -\frac{1}{K}$

Angle condition: $\angle G(s)H(s) = (2q + 1) \cdot 180^\circ, q = 0, \pm 1, \pm 2, \dots$

Magnitude condition: $|KG(s)H(s)| = 1$

4.1 Rules for Constructing Root Locus

Let $G(s)H(s) = \frac{K \prod (s + z_i)}{\prod (s + p_j)}$ have m zeros and n poles ($n \geq m$).

#	Rule	Statement
1	Number of branches	Equal to n (number of open-loop poles).
2	Start & End points	Locus starts ($K = 0$) at open-loop poles ; ends ($K \rightarrow \infty$) at open-loop zeros (finite or at ∞).
3	Symmetry	Root locus is always symmetric about the real axis .
4	Real axis segments	A point on the real axis lies on the locus if the total number of real poles and zeros to its right is odd .
5	Asymptotes	$(n - m)$ branches go to ∞ along asymptotes with angles: $\phi_k = \frac{(2k + 1) \cdot 180^\circ}{n - m}, k = 0, 1, \dots, (n - m - 1)$
6	Centroid of asymptotes	$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m}$ (intersection of asymptotes on real axis)
7	Breakaway / Break-in	Points where locus leaves/enters real axis. Found from $\frac{dK}{ds} = 0$, i.e. $\frac{d}{ds}[G(s)H(s)] = 0$.
8	$j\omega$-axis crossing	Substitute $s = j\omega$ in characteristic equation; solve for ω and K . (Or use Routh array.)
9	Angle of departure	From a complex pole p_k : $\phi_{dep} = 180^\circ - \sum \angle(p_k - z_i) + \sum \angle(p_k - p_j)$
10	Angle of arrival	At a complex zero z_k : $\phi_{arr} = 180^\circ + \sum \angle(z_k - p_j) - \sum \angle(z_k - z_i)$

4.2 Root Locus — Worked Example

Example: $G(s)H(s) = K/[s(s+2)(s+4)]$

Given: $n = 3$ poles at $s = 0, -2, -4$; $m = 0$ zeros.

Step 1 — Branches: 3 branches.

Step 2 — Real axis segments: Right of $s = 0$: 0 poles/zeros $\rightarrow$ not on locus. Between 0 and -2 : 1 pole to right $\rightarrow$ **on locus**. Between -2 and -4 : 2 poles to right $\rightarrow$ not on locus. Left of -4 : 3 poles to right $\rightarrow$ **on locus**.

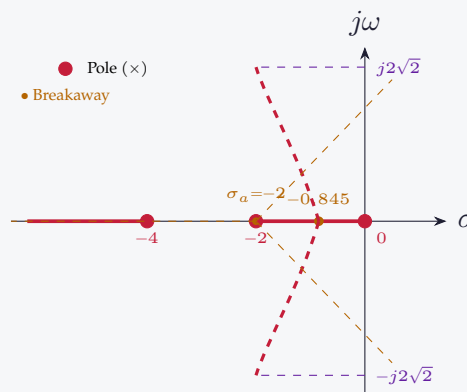
Step 3 — Asymptotes: $n - m = 3$, angles $= 60^\circ, 180^\circ, 300^\circ$.

Step 4 — Centroid: $\sigma_a = \frac{(0 - 2 - 4) - 0}{3} = -2$.

Step 5 — Breakaway: $\frac{dK}{ds} = 0 \Rightarrow 3s^2 + 12s + 8 = 0 \Rightarrow s \approx -0.845$ (valid, between 0 and -2).

Step 6 — $j\omega$ crossing: Characteristic eq: $s^3 + 6s^2 + 8s + K = 0$. Routh: $K_{max} = 48$, crossing at $\omega = \pm 2\sqrt{2}$.

Root Locus Sketch — $K/[s(s+2)(s+4)]$



Exam Tips — Root Locus

- **Locus starts at poles, ends at zeros** — memorise this first.
- **Real axis rule:** odd number of poles+zeros to the right $\rightarrow$ on locus.
- **Centroid formula:** $\sigma_a = (\sum p_i - \sum z_i)/(n - m)$ — very frequently asked.
- **Breakaway point:** solve $dK/ds = 0$ or equivalently $d[G(s)H(s)]/ds = 0$.
- **$j\omega$ crossing:** use Routh array or substitute $s = j\omega$ — gives K_{max} for stability.
- **Gain K at any point on locus:** $K = 1/|G(s)H(s)|$ at that s .

5. Compensator Design

A **compensator** (or controller) is added to the forward or feedback path to reshape the root locus so that the closed-loop poles are placed at desired locations, meeting the time-domain

specifications.

Compensator: A dynamic element $G_c(s)$ inserted in the control loop to improve transient response, steady-state accuracy, or both. The two fundamental types are **Lead** (improves transient) and **Lag** (improves steady-state).

5.1 Lead Compensator

Lead Compensator: A compensator whose **zero is closer to the origin than its pole**. It adds a net *positive* phase angle to the open-loop transfer function, effectively acting like a **PD controller**. It improves **transient response** — faster rise time, reduced overshoot.

Lead Compensator Transfer Function

$$G_c(s) = K_c \frac{s + z_c}{s + p_c}, \quad p_c > z_c > 0 \quad (\text{pole farther from origin than zero})$$

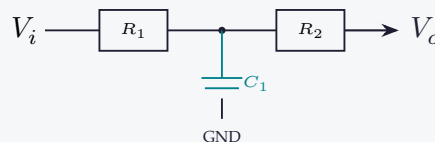
Phase contribution: $\phi = \angle(s + z_c) - \angle(s + p_c) > 0$ (positive phase lead)

Maximum phase lead: $\phi_{max} = \sin^{-1}\left(\frac{1 - \alpha}{1 + \alpha}\right)$, where $\alpha = z_c/p_c < 1$

Frequency of max lead: $\omega_{max} = \sqrt{z_c \cdot p_c}$

Op-Amp Realization of Lead Compensator

Lead Network (RC Circuit)



$$G_c(s) = \frac{R_2}{R_1 + R_2} \cdot \frac{1 + sR_1C_1}{1 + s\frac{R_1R_2}{R_1 + R_2}C_1}$$

Design Procedure (Root Locus)

Step 1: Determine desired closed-loop pole location s_d from specs (M_p, t_s, t_p) .

Step 2: Calculate the **angle deficiency**: $\phi = 180^\circ - \angle G(s_d)H(s_d)$.

Step 3: Place the compensator zero z_c and pole p_c such that they contribute angle ϕ at s_d .

Step 4: Determine gain K_c from the magnitude condition: $K_c = 1/|G_c(s_d)G(s_d)H(s_d)|$.

Step 5: Verify SSE and adjust if needed.

5.2 Lag Compensator

Lag Compensator: A compensator whose **pole is closer to the origin than its zero**. It adds a net *negative* phase angle (phase lag), effectively acting like a **PI controller**. It improves **steady-state accuracy** (increases low-frequency gain) without significantly altering the transient response.

Lag Compensator Transfer Function

$$G_c(s) = K_c \frac{s + z_c}{s + p_c}, \quad 0 < p_c < z_c \quad (\text{zero farther from origin than pole})$$

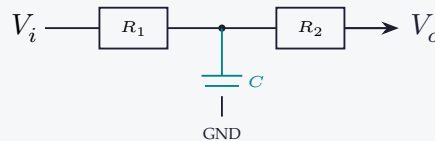
DC gain boost: $G_c(0) = K_c \frac{z_c}{p_c} = K_c \beta$, where $\beta = z_c/p_c > 1$

The lag compensator increases the **velocity constant** K_v by a factor of β , reducing SSE.

Placement rule: Place z_c and p_c very close to the origin (e.g., $z_c = 0.1$, $p_c = 0.01$) so they do not disturb the root locus significantly.

Op-Amp Realization of Lag Compensator

Lag Network (RC Circuit)



$$G_c(s) = \frac{1 + sR_2C}{1 + s(R_1 + R_2)C}, \quad \text{pole} < \text{zero}$$

Design Procedure (Root Locus)

- Step 1:** Design the uncompensated system to meet transient specs (find required K).
- Step 2:** Determine the factor $\beta = z_c/p_c$ needed to achieve the desired SSE improvement.
- Step 3:** Place z_c and p_c near the origin (e.g., $z_c = 0.1$, $p_c = z_c/\beta$).
- Step 4:** Verify that the lag compensator does not significantly shift the dominant poles.
- Step 5:** Recalculate gain K_c from magnitude condition at the desired pole.

5.3 Lead vs Lag — Comparison

Property	Lead Compensator	Lag Compensator
Pole-zero relation	$p_c > z_c$ (pole farther)	$p_c < z_c$ (pole closer)
Phase contribution	Positive (phase lead)	Negative (phase lag)
Equivalent to	PD controller	PI controller
Primary effect	Improves transient response	Improves steady-state accuracy
Bandwidth	Increases	Decreases slightly
Noise sensitivity	Higher (differentiating action)	Lower
Root locus effect	Shifts locus to the left (more stable)	Provides gain boost near origin
Typical use	Fast response needed	Low SSE needed without changing transient

5.4 Lead-Lag Compensator

Lead-Lag Compensator: Combines both lead and lag sections in cascade to simultaneously improve transient response *and* steady-state accuracy:

$$G_c(s) = K_c \frac{(s + z_1)(s + z_2)}{(s + p_1)(s + p_2)}, \quad p_1 > z_1 \text{ (lead)}, \quad p_2 < z_2 \text{ (lag)}$$

Equivalent to a **PID controller** in terms of effect.

Exam Tips — Compensators

- **Lead:** zero closer to origin than pole $\rightarrow$ phase lead $\rightarrow$ faster response.
- **Lag:** pole closer to origin than zero $\rightarrow$ phase lag $\rightarrow$ better SSE.
- **Angle deficiency** $= 180^\circ - \angle G(s_d)H(s_d)$ — must be provided by lead compensator.
- **β factor** for lag = ratio z_c/p_c = SSE improvement factor.
- Lead-lag = PID equivalent; used when both transient and SSE specs must be met.
- **RC lead network:** $G_c(s) \propto (1 + sR_1C)/(1 + s\frac{R_1R_2}{R_1+R_2}C)$ — pole $>$ zero.
- **RC lag network:** $G_c(s) \propto (1 + sR_2C)/(1 + s(R_1 + R_2)C)$ — pole $<$ zero.

Quick Revision — Module 3

Time Response, Root Locus & Compensators | EC601

Key Formulas at a Glance

Standard 2nd order TF	$G(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$
Poles	$s_{1,2} = -\zeta\omega_n \pm j\omega_d$
Damped freq.	$\omega_d = \omega_n \sqrt{1 - \zeta^2}$
Peak time	$t_p = \pi / \omega_d$
% Peak overshoot	$M_p = e^{-\pi\zeta / \sqrt{1 - \zeta^2}} \times 100\%$
Settling time (2%)	$t_s \approx 4 / (\zeta\omega_n)$
Settling time (5%)	$t_s \approx 3 / (\zeta\omega_n)$
Rise time (approx.)	$t_r \approx 1.8 / \omega_n$
SSE (step, Type 0)	$1 / (1 + K_p)$
SSE (ramp, Type 1)	$1 / K_v$
Asymptote angles	$\phi_k = (2k + 1) \cdot 180^\circ / (n - m)$
Centroid	$\sigma_a = (\sum p_i - \sum z_i) / (n - m)$
Gain at point s_d	$K = 1 / G(s_d)H(s_d) $
Lead TF	$K_c(s + z_c) / (s + p_c), p_c > z_c$
Lag TF	$K_c(s + z_c) / (s + p_c), p_c < z_c$
Max phase lead	$\phi_{max} = \sin^{-1}[(1 - \alpha) / (1 + \alpha)], \alpha = z_c / p_c$

One-Line Definitions

- ω_n : Natural frequency — speed of oscillation without damping.
- ζ : Damping ratio — controls oscillation; $\zeta = 1$ is critically damped.
- ω_d : Damped natural frequency — actual oscillation frequency in underdamped response.
- **Rise time t_r** : Time to go from 10% to 90% of final value.
- **Peak time t_p** : Time to reach first overshoot peak; $t_p = \pi / \omega_d$.
- **Overshoot M_p** : Maximum exceedance above final value; depends only on ζ .
- **Settling time t_s** : Time to stay within $\pm 2\%$ band; $t_s \approx 4 / (\zeta\omega_n)$.
- **Root locus**: Path of closed-loop poles as gain K varies $0 \rightarrow \infty$.

- **Breakaway point:** Point where locus leaves real axis; found from $dK/ds = 0$.
- **Centroid:** Intersection of asymptotes on real axis; $\sigma_a = (\sum p - \sum z)/(n - m)$.
- **Lead compensator:** Adds phase lead; improves transient; pole farther than zero.
- **Lag compensator:** Adds phase lag; improves SSE; pole closer than zero.
- **Angle deficiency:** Phase that lead compensator must supply at desired pole location.

Frequently Asked / Viva Questions

Q1. What is the standard second-order transfer function?

$$G(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2).$$

Q2. What is the effect of ζ on step response?

$\zeta < 1$: underdamped (oscillatory); $\zeta = 1$: critically damped (fastest non-oscillatory); $\zeta > 1$: overdamped (sluggish).

Q3. How is M_p related to ζ ?

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% \text{ — depends only on } \zeta, \text{ not } \omega_n.$$

Q4. What is the 2% settling time formula?

$$t_s \approx 4/(\zeta\omega_n).$$

Q5. What does the root locus start and end at?

Starts at open-loop poles ($K = 0$), ends at open-loop zeros ($K \rightarrow \infty$).

Q6. State the real axis rule for root locus.

A point on the real axis is on the locus if the total number of real poles and zeros to its right is odd.

Q7. What is the centroid formula?

$$\sigma_a = (\sum \text{poles} - \sum \text{zeros}) / (n - m).$$

Q8. How do you find the breakaway point?

Differentiate $K = -1/G(s)H(s)$ with respect to s and set to zero; or solve $d[G(s)H(s)]/ds = 0$.

Q9. What is the difference between lead and lag compensator?

Lead: $p > z$, adds positive phase, improves transient. Lag: $p < z$, adds negative phase, improves SSE.

Q10. What is angle deficiency?

The phase angle that the lead compensator must contribute at the desired closed-loop pole location: $\phi = 180^\circ - \angle G(s_d)H(s_d)$.

Frequently Asked / Viva Questions (contd.)

Q11. What is the β factor in lag compensator?

$\beta = z_c/p_c > 1$; it is the factor by which the velocity constant K_v (and hence SSE improvement) is multiplied.

Q12. What is a lead-lag compensator equivalent to?

A PID controller — provides both transient improvement (lead) and SSE improvement (lag).

Damping Cases — Quick Reference

ζ value	Pole type	Response
$\zeta = 0$	$\pm j\omega_n$	Undamped — sustained oscillation
$0 < \zeta < 1$	Complex conjugate	Underdamped — decaying oscillation
$\zeta = 1$	Two equal real	Critically damped — fastest non-oscillatory
$\zeta > 1$	Two distinct real	Overdamped — no oscillation, slow

Root Locus Rules — Quick Reference

#	Rule	Formula / Statement
1	Branches	n branches (= no. of poles)
2	Start/End	$K = 0$: poles; $K \rightarrow \infty$: zeros
3	Symmetry	Symmetric about real axis
4	Real axis	Odd poles+zeros to the right
5	Asymptote angles	$(2k + 1) \cdot 180^\circ / (n - m)$
6	Centroid	$(\sum p_i - \sum z_i) / (n - m)$
7	Breakaway	$dK/ds = 0$
8	$j\omega$ crossing	$s = j\omega$ in char. eq. or Routh

Module 4

Frequency Response Analysis — Introduction

1. Frequency Response Analysis — Introduction

Frequency response analysis studies how a system responds to sinusoidal inputs of varying frequency. For a stable LTI system with transfer function $G(j\omega)$, a sinusoidal input $r(t) = A \sin(\omega t)$ produces a steady-state output $c(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega))$.

Frequency Response Function: Obtained by substituting $s = j\omega$ in the transfer function:

$$G(j\omega) = G(s)|_{s=j\omega} = |G(j\omega)| \angle \phi(\omega)$$

Magnitude: $|G(j\omega)|$ — ratio of output to input amplitude.

Phase: $\phi(\omega) = \angle G(j\omega)$ — phase shift of output relative to input.

Plot Type	Description
Polar Plot	Locus of $G(j\omega)$ in the complex plane as $\omega : 0 \rightarrow \infty$. Also called Nyquist plot (open-loop).
Bode Plot	Two separate plots: $ G(j\omega) $ in dB vs $\log \omega$, and $\angle G(j\omega)$ in degrees vs $\log \omega$.
Nyquist Plot	Polar plot of $G(j\omega)H(j\omega)$ for $\omega : -\infty \rightarrow +\infty$; used for Nyquist stability criterion.
Nichols Chart	Magnitude vs phase plot; used for closed-loop frequency response from open-loop data.

2. Polar Plots

Polar Plot: A plot of $G(j\omega)$ in the complex (Re–Im) plane as ω varies from 0 to $+\infty$. Each point on the curve represents the tip of the phasor $G(j\omega)$ at a particular frequency. The distance from the origin is $|G(j\omega)|$ and the angle is $\angle G(j\omega)$.

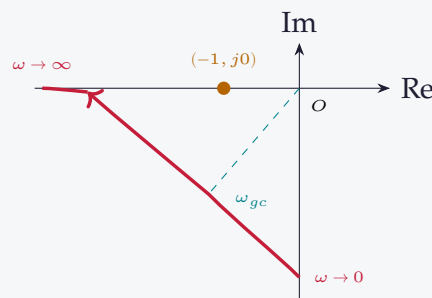
2.1 Polar Plot of Standard Factors

Factor	At $\omega = 0$	At $\omega \rightarrow \infty$
K (gain)	$(K, 0^\circ)$	$(K, 0^\circ)$ — point on real axis
$j\omega$ (integrator)	$\infty \angle -90^\circ$	$0 \angle -90^\circ$ — negative imaginary axis
$1/j\omega$	$\infty \angle -90^\circ$	$0 \angle -90^\circ$ — negative imaginary axis
$1/(1 + j\omega T)$	$(1, 0^\circ)$	$(0, -90^\circ)$ — 4th quadrant semicircle
$1/(j\omega)^2$	$\infty \angle -180^\circ$	$0 \angle -180^\circ$ — negative real axis
$1/[j\omega(1 + j\omega T)]$	$\infty \angle -90^\circ$	$0 \angle -180^\circ$ — 3rd quadrant

2.2 Key Properties of Polar Plots

- The **starting point** ($\omega = 0$) and **ending point** ($\omega \rightarrow \infty$) are determined by the system type and order.
- For a **Type 0** system: starts on positive real axis; for **Type 1**: starts at -90° infinity; for **Type 2**: starts at -180° infinity.
- The **number of clockwise encirclements** of the $(-1, j0)$ point determines closed-loop stability (Nyquist criterion).
- The **gain crossover frequency** ω_{gc} : where $|G(j\omega)| = 1$ (0 dB).
- The **phase crossover frequency** ω_{pc} : where $\angle G(j\omega) = -180^\circ$.

Polar Plot — Type 1, Second Order: $G(j\omega) = K/[j\omega(1 + j\omega T)]$



3. Bode Plot

Bode Plot: Two semi-log plots of the open-loop frequency response $G(j\omega)$:

- **Magnitude plot:** $20 \log_{10} |G(j\omega)|$ in dB vs $\log_{10} \omega$ (log scale).
- **Phase plot:** $\angle G(j\omega)$ in degrees vs $\log_{10} \omega$ (log scale).

Bode plots use **asymptotic approximations** (straight-line segments) for quick

sketching.

3.1 Standard Factors and Their Bode Contributions

Factor	Magnitude (dB)	Phase
Gain K	$20 \log K$ (constant)	0° (if $K > 0$); -180° (if $K < 0$)
$j\omega$ (zero at origin)	$+20$ dB/dec, passes through 0 dB at $\omega = 1$	$+90^\circ$ (constant)
$1/j\omega$ (pole at origin)	-20 dB/dec, passes through 0 dB at $\omega = 1$	-90° (constant)
$(1 + j\omega T)$ (simple zero)	0 dB for $\omega \ll 1/T$; $+20$ dB/dec for $\omega \gg 1/T$	$0^\circ \rightarrow +90^\circ$, 45° at $\omega = 1/T$
$1/(1 + j\omega T)$ (simple pole)	0 dB for $\omega \ll 1/T$; -20 dB/dec for $\omega \gg 1/T$	$0^\circ \rightarrow -90^\circ$, -45° at $\omega = 1/T$
$1/(j\omega)^2$ (double pole)	-40 dB/dec	-180° (constant)
2nd order factor	0 dB below ω_n ; -40 dB/dec above ω_n ; resonance peak near ω_n	$0^\circ \rightarrow -180^\circ$, -90° at ω_n

3.2 Corner Frequency (Break Frequency)

Corner Frequency

For a factor $(1 + j\omega T)$: **corner frequency** $\omega_c = 1/T$ rad/s.

Asymptotic error at corner frequency:

- Magnitude: actual value differs from asymptote by ± 3 dB at $\omega = \omega_c$.
- Phase: actual = $\pm 45^\circ$ at ω_c ; asymptote approximation has $\pm 5.7^\circ$ max error.

3.3 Bode Plot Sketching Procedure

- Step 1:** Write $G(j\omega)$ in **time-constant form**: factor out constants, express each factor as $(1 + j\omega T)$.
- Step 2:** Compute the **initial slope**: $-20N$ dB/dec where N = system type (number of poles at origin).
- Step 3:** Mark all **corner frequencies** $\omega_c = 1/T$ on the ω -axis.
- Step 4:** Draw the **magnitude asymptotes**: slope changes by $+20$ dB/dec at each zero corner, -20 dB/dec at each pole corner.
- Step 5:** Apply **3 dB corrections** at each corner frequency if accuracy is needed.
- Step 6:** Draw the **phase plot**: sum contributions from each factor; use $\pm 45^\circ$ /decade approximation.

Example: $G(j\omega) = \frac{10}{j\omega(1 + 0.1j\omega)(1 + 0.5j\omega)}$

Factors: Gain $K = 10$ (+20 dB), pole at origin (−20 dB/dec from start), poles at $\omega_c = 10$ and $\omega_c = 2$.

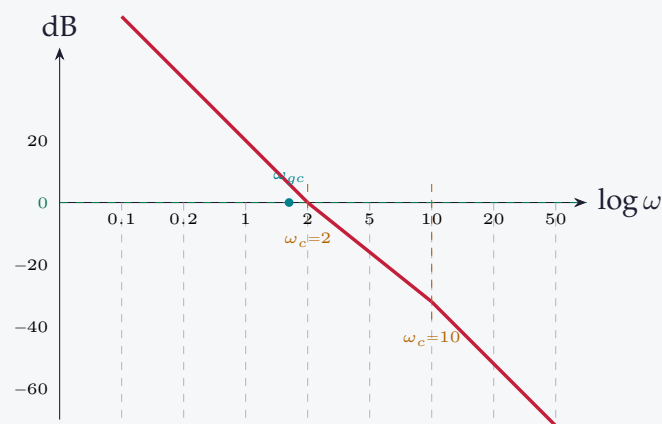
Initial value: At $\omega = 1$: $20 \log(10/1) = 20$ dB (from gain and pole at origin).

Slope changes:

- $\omega < 2$: slope = −20 dB/dec (one pole at origin)
- $2 < \omega < 10$: slope = −40 dB/dec (pole at $\omega = 2$ adds −20)
- $\omega > 10$: slope = −60 dB/dec (pole at $\omega = 10$ adds another −20)

Phase: -90° (pole at origin) $-\tan^{-1}(0.5\omega) - \tan^{-1}(0.1\omega)$

Bode Magnitude Plot Sketch



4. Stability in Frequency Domain

4.1 Gain Margin (GM)

Gain Margin: The factor by which the open-loop gain can be *increased* before the closed-loop system becomes unstable. Measured at the **phase crossover frequency** ω_{pc} (where $\angle G(j\omega)H(j\omega) = -180^\circ$).

$$GM = \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|}$$

In dB: $GM_{dB} = -20 \log_{10} |G(j\omega_{pc})H(j\omega_{pc})|$

Stable system: $GM > 1$ (i.e., $GM_{dB} > 0$ dB). **Unstable:** $GM < 1$ ($GM_{dB} < 0$ dB).

4.2 Phase Margin (PM)

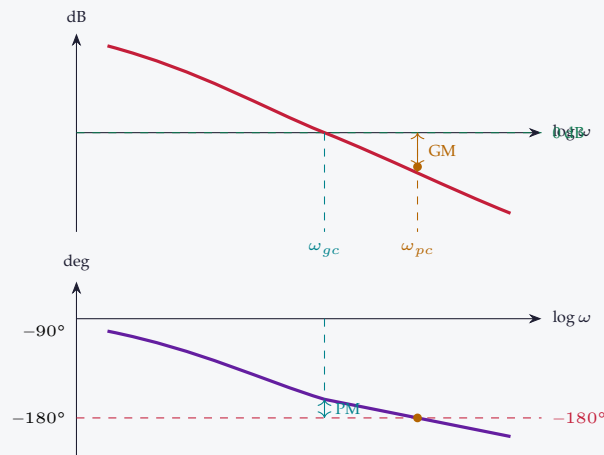
Phase Margin: The additional phase lag that can be introduced before the closed-loop system becomes unstable. Measured at the **gain crossover frequency** ω_{gc} (where $|G(j\omega)H(j\omega)| = 1$, i.e., 0 dB).

$$PM = 180^\circ + \angle G(j\omega_{gc})H(j\omega_{gc})$$

Stable system: $PM > 0^\circ$. **Unstable:** $PM < 0^\circ$.

Typical design target: $GM \geq 6$ dB, $PM \geq 30^\circ$ – 60° .

GM and PM on Bode Plot



Margin	Definition	Measurement Point
Gain Margin (GM)	$1/ G(j\omega_{pc}) $	At phase crossover freq ω_{pc} ($\angle = -180^\circ$)
Phase Margin (PM)	$180^\circ + \angle G(j\omega_{gc})$	At gain crossover freq ω_{gc} ($ G = 1$)
Stable condition	$GM > 1$ (> 0 dB)	$PM > 0^\circ$
Good design	$GM \geq 6$ dB	$PM = 30^\circ$ – 60°

Exam Tips — GM and PM

- **GM** is read from the magnitude plot at ω_{pc} ; **PM** is read from the phase plot at ω_{gc} .
- If the magnitude curve *never* crosses 0 dB, ω_{gc} does not exist — system is always stable for that gain.
- If the phase curve *never* reaches -180° , ω_{pc} does not exist — $GM = \infty$.
- **Higher PM** $\Rightarrow$ less overshoot, more damping. $PM \approx 100\zeta$ for $\zeta < 0.7$.
- **Conditionally stable systems** can have $GM > 0$ dB but become unstable if gain is *reduced*.

5. Nyquist Plot

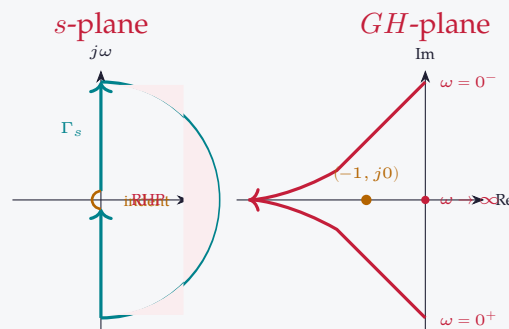
Nyquist Plot: The polar plot of $G(j\omega)H(j\omega)$ for ω varying from $-\infty$ to $+\infty$ (i.e., the complete Nyquist contour). It is the mirror image of the polar plot about the real axis for negative frequencies. Used to apply the **Nyquist Stability Criterion**.

5.1 Nyquist Contour

The **Nyquist contour** Γ_s in the s -plane encloses the entire right half-plane (RHP):

- Segment 1: $s = j\omega, \omega : 0^+ \rightarrow +\infty$ (positive imaginary axis)
- Segment 2: $s = Re^{j\theta}, R \rightarrow \infty, \theta : +90^\circ \rightarrow -90^\circ$ (large semicircle)
- Segment 3: $s = j\omega, \omega : -\infty \rightarrow 0^-$ (negative imaginary axis)
- **Indentation:** Small semicircle of radius $\varepsilon \rightarrow 0$ around any poles on the imaginary axis.

Nyquist Contour in s -plane and Corresponding $G(j\omega)H(j\omega)$ Plot



6. Nyquist Stability Criterion

Nyquist Stability Criterion: Based on the **Principle of the Argument** (Cauchy's theorem). For a closed-loop system with open-loop transfer function $G(s)H(s)$:

$$N = Z - P$$

where:

- N = number of **clockwise encirclements** of the $(-1, j0)$ point by the Nyquist plot of $G(j\omega)H(j\omega)$.
- Z = number of **closed-loop poles** in the RHP (zeros of $1 + GH$).
- P = number of **open-loop poles** of $G(s)H(s)$ in the RHP.

For stability: $Z = 0$, therefore $N = -P$ (i.e., P counter-clockwise encirclements if $P \neq 0$; no encirclements if $P = 0$).

6.1 Applying the Nyquist Criterion — Step by Step

Step 1: Determine P = number of open-loop poles of $G(s)H(s)$ in the RHP.

Step 2: Sketch the Nyquist plot of $G(j\omega)H(j\omega)$ for $\omega : 0^+ \rightarrow +\infty$.

Step 3: Mirror the plot about the real axis for $\omega : -\infty \rightarrow 0^-$.

Step 4: Handle poles on the $j\omega$ -axis by indenting the contour (small semicircle of radius $\varepsilon \rightarrow 0$).

Step 5: Count N = net clockwise encirclements of $(-1, j0)$.

Step 6: Compute $Z = N + P$. If $Z = 0$: **stable**. If $Z > 0$: **unstable** (Z RHP poles).

Example: $G(s)H(s) = K/[s(s+1)(s+2)]$, find stability range

Open-loop poles: $s = 0$ (on $j\omega$ -axis), $s = -1$, $s = -2$ (all in LHP). So $P = 0$.

Nyquist plot crosses the negative real axis at ω_{pc} where $\angle G(j\omega)H(j\omega) = -180^\circ$.

At ω_{pc} : $\angle = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(\omega/2) = -180^\circ \Rightarrow \tan^{-1}(\omega) + \tan^{-1}(\omega/2) = 90^\circ$
 $\Rightarrow \omega_{pc} = \sqrt{2}$ rad/s.

$$|G(j\sqrt{2})H(j\sqrt{2})| = \frac{K}{\sqrt{2} \cdot \sqrt{3} \cdot \sqrt{4}} = \frac{K}{2\sqrt{6}}$$

For stability: $N = 0$ requires the plot does not encircle $(-1, j0)$:

$$\frac{K}{2\sqrt{6}} < 1 \Rightarrow K < 2\sqrt{6} \approx 4.9$$

Gain Margin: $GM = 2\sqrt{6}/K$. For $K = 1$: $GM = 2\sqrt{6} \approx 4.9$ (13.8 dB).

6.2 Simplified Nyquist Criterion (Minimum Phase Systems)

For Minimum Phase Systems ($P = 0$)

A minimum phase system (no open-loop RHP poles or zeros) is **closed-loop stable** if and only if the Nyquist plot of $G(j\omega)H(j\omega)$ does **not encircle** the $(-1, j0)$ point.

Equivalently (from Bode plot): **stable** if $|G(j\omega_{pc})H(j\omega_{pc})| < 1$, i.e., $GM > 0$ dB AND $PM > 0^\circ$.

Exam Tips — Nyquist Criterion

- $N = Z - P$ — memorise this. $Z = 0$ for stability.
- For **minimum phase** systems ($P = 0$): stable iff no encirclement of $(-1, j0)$.
- **Clockwise** encirclement = positive N ; **counter-clockwise** = negative N .
- Poles on $j\omega$ -axis (e.g., integrators): indent contour to the right; the large semicircle maps to origin.
- The Nyquist plot is the **mirror image** of the polar plot about the real axis.
- $(-1, j0)$ **point** is the critical point — corresponds to $|GH| = 1$ and $\angle GH = -180^\circ$.

7. Performance Specifications in Frequency Domain

Frequency Domain Specifications describe the closed-loop system's behaviour in terms of its frequency response $T(j\omega) = C(j\omega)/R(j\omega)$. They are directly related to time-domain performance.

7.1 Closed-Loop Frequency Response Specifications

Specification	Symbol	Definition
Resonant Peak	M_r	Maximum value of $ T(j\omega) $. Indicates relative stability; high M_r means low damping.
Resonant Frequency	ω_r	Frequency at which $ T(j\omega) $ is maximum.
Bandwidth	ω_{BW}	Frequency at which $ T(j\omega) $ drops to $1/\sqrt{2}$ (−3 dB) of its DC value. Indicates speed of response.
Cutoff Rate	—	Rate at which $ T(j\omega) $ decreases beyond ω_{BW} ; indicates noise rejection.

7.2 Formulas for Second-Order System

Closed-Loop Frequency Response — 2nd Order

For $T(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$:

$$|T(j\omega)| = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}}, \quad u = \omega/\omega_n$$

Resonant frequency: $\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$ (exists only for $\zeta < 1/\sqrt{2} \approx 0.707$)

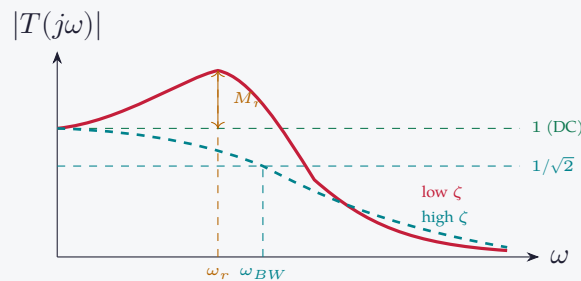
Resonant peak: $M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$ (for $\zeta < 0.707$)

Bandwidth: $\omega_{BW} = \omega_n \sqrt{(1 - 2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$

7.3 Relation Between Frequency and Time Domain Specs

Freq. Domain	Relation	Time Domain Equivalent
High ω_{BW}	$\omega_{BW} \propto \omega_n$	Faster rise time, faster response
High M_r	$M_r \uparrow$ as $\zeta \downarrow$	High overshoot M_p
$M_r = 1$	$\zeta \geq 0.707$	No overshoot (overdamped/critically)
Phase Margin	$PM \approx 100\zeta$ (for $\zeta < 0.7$)	Damping ratio ζ
Gain Margin	Larger GM	More stable, less sensitive to gain
ω_{gc}	$\omega_{gc} \approx \omega_{BW}$	Bandwidth approximation

Closed-Loop Frequency Response (Resonance Peak)



Exam Tips — Frequency Domain Specs

- $M_r > 1.3$ (about 2.3 dB) indicates poor relative stability (excessive overshoot).
- $\omega_r < \omega_n$ always (resonant frequency is less than natural frequency).
- $\omega_{BW} > \omega_n$ always (bandwidth exceeds natural frequency).
- $PM \approx 100\zeta$ for $\zeta < 0.7$ — very useful approximation in design.
- **Bandwidth** determines speed: wider BW = faster response but more noise.
- For $\zeta \geq 0.707$: $M_r = 1$ (no resonance peak); response is monotonically decreasing.

8. Frequency Response — Concept Mappings & Summary

Frequency response analysis involves several interrelated graphical methods and specifications. The mappings below summarize key relationships that resolve common areas of confusion.

1. Crossover Frequencies & Bode Plot Stability

Crossover frequencies are the measurement points for stability margins. GM is measured at the phase crossover frequency ω_{pc} , and PM is measured at the gain crossover frequency ω_{gc} .

Parameter / Case	Stable System	Marginally Stable	Unstable System
Crossover Freqs	$\omega_{gc} < \omega_{pc}$	$\omega_{gc} = \omega_{pc}$	$\omega_{gc} > \omega_{pc}$
Gain Margin (GM_{dB})	Positive (> 0 dB)	Zero ($= 0$ dB)	Negative (< 0 dB)
Phase Margin (PM)	Positive ($> 0^\circ$)	Zero ($= 0^\circ$)	Negative ($< 0^\circ$)

Critical Rules for Crossover Measurements

- **Gain Margin (GM):** Read from the Magnitude plot at $\omega = \omega_{pc}$ (where Phase $= -180^\circ$).
- **Phase Margin (PM):** Read from the Phase plot at $\omega = \omega_{gc}$ (where Magnitude $= 0$ dB).

2. Nyquist Stability Criterion Mapping

The closed-loop stability depends on the number of open-loop RHP poles (P) and the net clockwise encirclements (N) of the critical point $(-1, j0)$ by the Nyquist path:

$$\text{For Closed-Loop Stability: } Z = N + P = 0 \implies N = -P$$

- **Stable Open-Loop ($P = 0$):** Requires $N = 0$. The Nyquist plot must not encircle $(-1, j0)$.
- **Unstable Open-Loop ($P > 0$):** Requires $N = -P$. The Nyquist plot must encircle $(-1, j0)$ exactly P times in the **counter-clockwise** direction.

3. Time-Domain vs. Frequency-Domain Specifications

Closed-loop frequency response specifications map directly to transient and steady-state time-domain performance.

Frequency Domain Spec	Physical Meaning	Time Domain Equivalent
Resonant Peak M_r	Peak overshoot in frequency domain. High M_r indicates low damping ratio ζ .	Peak Overshoot M_p in transient response. (Large $M_r \leftrightarrow$ large M_p).
Resonant Freq ω_r	Frequency at which peak magnitude occurs. $\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$.	Damped natural frequency $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. (Usually $\omega_r < \omega_d < \omega_n$).
Bandwidth ω_{BW}	Range of frequencies system can track. Indication of speed.	Speed of response. Larger $\omega_{BW} \implies$ smaller rise time t_r and settling time t_s .
Phase Margin PM	Phase lag safety margin at unit gain. $PM \approx 100\zeta$.	Damping ratio ζ . Higher $PM \implies$ higher damping (less oscillatory).

Quick Revision — Module 4

Frequency Response Analysis | EC601

Key Formulas at a Glance

Frequency response	$G(j\omega) = G(s) _{s=j\omega}$
Gain Margin	$GM = 1/ G(j\omega_{pc})H(j\omega_{pc}) $
GM in dB	$GM_{dB} = -20 \log G(j\omega_{pc}) $
Phase Margin	$PM = 180^\circ + \angle G(j\omega_{gc})H(j\omega_{gc})$
Phase crossover freq	ω_{pc} : where $\angle GH = -180^\circ$
Gain crossover freq	ω_{gc} : where $ GH = 1$ (0 dB)
Nyquist criterion	$N = Z - P$; stable if $Z = 0$
Resonant peak (2nd order)	$M_r = 1/(2\zeta\sqrt{1-\zeta^2})$, $\zeta < 0.707$
Resonant frequency	$\omega_r = \omega_n \sqrt{1-2\zeta^2}$
Bandwidth (2nd order)	$\omega_{BW} = \omega_n \sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$
PM–damping relation	$PM \approx 100\zeta$ (for $\zeta < 0.7$)
Bode: simple pole slope	-20 dB/dec above corner freq $\omega_c = 1/T$
Bode: double pole slope	-40 dB/dec above ω_n
Bode: phase at corner	-45° at ω_c for simple pole

One-Line Definitions

- **Frequency Response:** System's steady-state response to sinusoidal input; $G(j\omega)$.
- **Polar Plot:** Locus of $G(j\omega)$ in complex plane as $\omega : 0 \rightarrow \infty$.
- **Bode Plot:** Semi-log plots of magnitude (dB) and phase (deg) vs $\log \omega$.
- **Corner Frequency:** $\omega_c = 1/T$; slope changes by ± 20 dB/dec here.
- **Gain Crossover Freq ω_{gc} :** Where $|GH| = 1$ (0 dB); PM is measured here.
- **Phase Crossover Freq ω_{pc} :** Where $\angle GH = -180^\circ$; GM is measured here.
- **Gain Margin:** Factor by which gain can increase before instability; $GM > 0$ dB for stability.
- **Phase Margin:** Additional phase lag before instability; $PM > 0^\circ$ for stability.
- **Nyquist Plot:** Polar plot of $GH(j\omega)$ for $\omega : -\infty \rightarrow +\infty$.
- **Nyquist Criterion:** $N = Z - P$; stable if $Z = 0$ (no RHP closed-loop poles).

- **Resonant Peak M_r :** Maximum of $|T(j\omega)|$; indicates damping ($M_r \uparrow$ as $\zeta \downarrow$).
- **Bandwidth ω_{BW} :** Freq where $|T|$ drops to -3 dB; indicates response speed.
- **Minimum Phase System:** No RHP poles or zeros; $P = 0$; stable iff no encirclement of $(-1, j0)$.

Frequently Asked / Viva Questions

Q1. What is the frequency response of a system?

The steady-state output-to-input ratio for sinusoidal inputs; obtained by substituting $s = j\omega$ in $G(s)$.

Q2. What is the gain crossover frequency?

The frequency ω_{gc} where $|G(j\omega)H(j\omega)| = 1$ (0 dB). Phase margin is measured here.

Q3. What is the phase crossover frequency?

The frequency ω_{pc} where $\angle G(j\omega)H(j\omega) = -180^\circ$. Gain margin is measured here.

Q4. Define Gain Margin.

$GM = 1/|GH(j\omega_{pc})|$. It is the factor by which gain can be increased before instability. Stable if $GM > 1$ (0 dB).

Q5. Define Phase Margin.

$PM = 180^\circ + \angle GH(j\omega_{gc})$. Additional phase lag before instability. Stable if $PM > 0^\circ$.

Q6. State the Nyquist stability criterion.

$N = Z - P$. For stability $Z = 0$, so $N = -P$. For minimum phase ($P = 0$): no encirclement of $(-1, j0)$.

Q7. What is N, Z, P in Nyquist criterion?

N = clockwise encirclements of $(-1, j0)$; Z = RHP closed-loop poles; P = RHP open-loop poles.

Q8. What is the slope of Bode magnitude plot for a simple pole?

-20 dB/decade above the corner frequency $\omega_c = 1/T$.

Q9. What is the resonant peak M_r ?

Maximum value of closed-loop frequency response; $M_r = 1/(2\zeta\sqrt{1-\zeta^2})$ for 2nd order. Exists only for $\zeta < 0.707$.

Q10. What is bandwidth?

Frequency at which $|T(j\omega)|$ drops to $1/\sqrt{2}$ (-3 dB) of DC value. Wider BW = faster response.

Frequently Asked / Viva Questions (contd.)

Q11. Relation between PM and damping ratio?

$PM \approx 100\zeta$ for $\zeta < 0.7$. So $PM = 45^\circ$ corresponds to $\zeta \approx 0.45$.

Q12. What is a minimum phase system?

A system with no RHP poles or zeros and no time delay. Its Bode magnitude uniquely determines the phase.

Stability Summary

Condition	Stable	Unstable
Gain Margin	$GM > 1$ (> 0 dB)	$GM < 1$ (< 0 dB)
Phase Margin	$PM > 0^\circ$	$PM < 0^\circ$
Nyquist ($P = 0$)	No encirclement of $(-1, j0)$	Encircles $(-1, j0)$
Nyquist (general)	$Z = N + P = 0$	$Z = N + P > 0$
Bode (min. phase)	$\omega_{gc} < \omega_{pc}$	$\omega_{gc} > \omega_{pc}$

Module 5

State Variable Analysis — Introduction

1. State Variable Analysis — Introduction

Classical control (transfer function approach) describes a system only in terms of its input-output relationship and is limited to LTI, SISO systems. **State variable analysis** (modern control) provides a complete internal description of the system, handles MIMO systems, time-varying systems, and nonlinear systems, and is directly suited for computer-based design.

State: The state of a system at time t_0 is the minimum set of variables (state variables) that, together with the input for $t \geq t_0$, completely determines the future behaviour of the system.

State Variable: Each member of the minimum set of variables that completely describes the state of the system. For an n -th order system, there are exactly n state variables.

State Vector: The column vector $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T$ formed by all n state variables.

State Space: The n -dimensional space whose axes are the state variables $x_1, x_2, \dots, x_n$. Each point in state space represents a unique state of the system.

1.1 Why State Variables?

Aspect	Transfer Function	State Space
System type	SISO, LTI only	MIMO, nonlinear, time-varying
Internal info	No (black box)	Yes (complete internal description)
Initial conditions	Difficult to handle	Naturally included
Computer design	Indirect	Direct (matrix methods)
Stability analysis	Characteristic equation	Eigenvalues of A matrix
Design methods	Root locus, Bode	Pole placement, optimal control

2. State Space Representation

Standard State Space Form

State equation:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

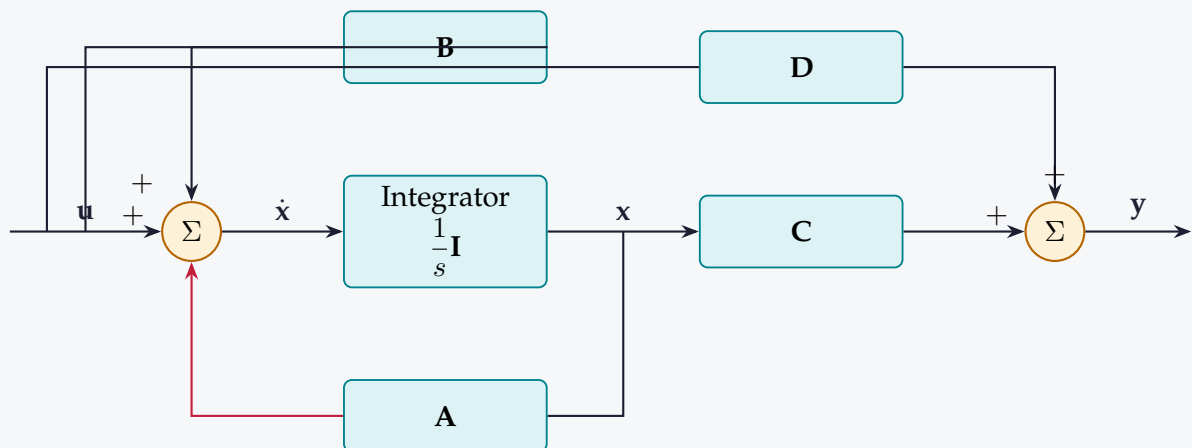
Output equation:

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

where:

- $\mathbf{x}(t) \in \mathbb{R}^n$ — state vector (n state variables)
- $\mathbf{u}(t) \in \mathbb{R}^r$ — input vector (r inputs)
- $\mathbf{y}(t) \in \mathbb{R}^m$ — output vector (m outputs)
- $\mathbf{A}$ ($n \times n$) — system/plant matrix (dynamics)
- $\mathbf{B}$ ($n \times r$) — input matrix
- $\mathbf{C}$ ($m \times n$) — output matrix
- $\mathbf{D}$ ($m \times r$) — direct transmission (feedthrough) matrix

State Space Block Diagram



2.1 Choosing State Variables

For a physical system, state variables are typically the **energy storage variables**:

- **Electrical circuits:** Capacitor voltages v_C and inductor currents i_L
- **Mechanical systems:** Displacement x and velocity $\dot{x}$
- **Thermal systems:** Temperatures at each node

For an n -th order ODE, the standard choice is the **phase variable form**: $x_1 = y$, $x_2 = \dot{y}$, ..., $x_n = y^{(n-1)}$.

2.2 Phase Variable Form — Example

Example: 3rd Order System

Given ODE: $\ddot{y} + a_2\dot{y} + a_1\dot{y} + a_0y = b_0u$

State variables: $x_1 = y, \quad x_2 = \dot{y}, \quad x_3 = \ddot{y}$

State equations:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -a_0x_1 - a_1x_2 - a_2x_3 + b_0u$$

Matrix form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ b_0 \end{bmatrix}}_{\mathbf{B}} u, \quad y = \underbrace{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}_{\mathbf{C}} \mathbf{x}$$

The **A** matrix in this form is called the **companion matrix** or **phase variable canonical form**.

3. State Transition Matrix (STM)

State Transition Matrix (STM) $\Phi(t)$: Also called the **matrix exponential** $e^{\mathbf{A}t}$. It describes how the state vector transitions from its initial value $\mathbf{x}(0)$ to $\mathbf{x}(t)$ in the absence of any input (homogeneous case):

$$\mathbf{x}(t) = \Phi(t) \mathbf{x}(0) = e^{\mathbf{A}t} \mathbf{x}(0)$$

3.1 Definition and Computation

Matrix Exponential — Definition

$$\Phi(t) = e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2t^2}{2!} + \frac{\mathbf{A}^3t^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!}$$

Methods to compute $e^{\mathbf{A}t}$:

1. **Infinite series** (above) — impractical for large n .
2. **Laplace transform method:** $\Phi(t) = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$
3. **Cayley-Hamilton theorem:** Express $e^{\mathbf{A}t}$ as a polynomial in $\mathbf{A}$ of degree $\leq n - 1$.
4. **Diagonalisation:** If $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$, then $e^{\mathbf{A}t} = \mathbf{P}e^{\mathbf{\Lambda}t}\mathbf{P}^{-1}$.

Laplace Method (Most Exam-Relevant)

$$\Phi(t) = e^{\mathbf{A}t} = \mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A})^{-1} \}$$

Steps:

1. Form $(s\mathbf{I} - \mathbf{A})$ by subtracting $\mathbf{A}$ from s times the identity matrix.
2. Compute the inverse: $(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}$
3. Take the inverse Laplace transform of each element.

3.2 Properties of the STM

#	Property	Expression
1	Initial value	$\Phi(0) = \mathbf{I}$ (identity matrix)
2	Inverse	$\Phi^{-1}(t) = \Phi(-t) = e^{-\mathbf{A}t}$
3	Composition	$\Phi(t_1 + t_2) = \Phi(t_1) \Phi(t_2)$
4	Derivative	$\dot{\Phi}(t) = \mathbf{A} \Phi(t) = \Phi(t) \mathbf{A}$
5	Transition	$\Phi(t_2 - t_1) = \Phi(t_2) \Phi^{-1}(t_1)$
6	Determinant	$\det[\Phi(t)] = e^{\text{tr}(\mathbf{A}) \cdot t} \neq 0$ (always non-singular)

Example: Compute STM for $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$\det(s\mathbf{I} - \mathbf{A}) = s(s+3) + 2 = s^2 + 3s + 2 = (s+1)(s+2)$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

Using partial fractions and inverse Laplace:

$$\Phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

4. Solution of State Equations**4.1 Homogeneous State Equation (Zero Input)**

When $\mathbf{u}(t) = \mathbf{0}$, the state equation reduces to:

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t)$$

Solution — Homogeneous (Zero Input)

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}(0) = \Phi(t) \mathbf{x}(0)$$

This is the **free response** or **zero-input response**. It depends only on the initial state $\mathbf{x}(0)$ and the system matrix $\mathbf{A}$.

The output is: $\mathbf{y}(t) = \mathbf{C} \Phi(t) \mathbf{x}(0)$

Derivation: Assume $\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{c}$. Then $\dot{\mathbf{x}} = \mathbf{A}e^{\mathbf{A}t} \mathbf{c} = \mathbf{A} \mathbf{x}$. At $t = 0$: $\mathbf{x}(0) = \mathbf{c}$. Hence $\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}(0)$.

4.2 Non-Homogeneous State Equation (With Input)

When $\mathbf{u}(t) \neq 0$:

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t)$$

Solution — Non-Homogeneous (With Input)

$$\mathbf{x}(t) = \underbrace{e^{\mathbf{A}t} \mathbf{x}(0)}_{\text{zero-input}} + \underbrace{\int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau}_{\text{zero-state (convolution)}}$$

$$= \Phi(t) \mathbf{x}(0) + \int_0^t \Phi(t - \tau) \mathbf{B} \mathbf{u}(\tau) d\tau$$

Output:

$$\mathbf{y}(t) = \mathbf{C} \Phi(t) \mathbf{x}(0) + \int_0^t \mathbf{C} \Phi(t - \tau) \mathbf{B} \mathbf{u}(\tau) d\tau + \mathbf{D} \mathbf{u}(t)$$

Derivation (variation of parameters): Multiply both sides of $\dot{\mathbf{x}} - \mathbf{A} \mathbf{x} = \mathbf{B} \mathbf{u}$ by the integrating factor $e^{-\mathbf{A}t}$:

$$\frac{d}{dt} [e^{-\mathbf{A}t} \mathbf{x}(t)] = e^{-\mathbf{A}t} \mathbf{B} \mathbf{u}(t)$$

Integrating from 0 to t and multiplying by $e^{\mathbf{A}t}$ gives the result above.

4.3 Laplace Domain Solution

Taking the Laplace transform of $\dot{\mathbf{x}} = \mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{u}$:

Laplace Domain Solution

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{x}(0) + \mathbf{B}\mathbf{U}(s)$$

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B}\mathbf{U}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}(0) + [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}] \mathbf{U}(s)$$

5. Transfer Function from State Space

The **transfer function matrix** $\mathbf{G}(s)$ relates the output $\mathbf{Y}(s)$ to the input $\mathbf{U}(s)$ with zero initial conditions $[\mathbf{x}(0) = \mathbf{0}]$:

$$\mathbf{G}(s) = \left. \frac{\mathbf{Y}(s)}{\mathbf{U}(s)} \right|_{\mathbf{x}(0)=\mathbf{0}}$$

Transfer Function from State Space (Zero I.C.)

Setting $\mathbf{x}(0) = \mathbf{0}$ in the Laplace solution:

$$\mathbf{Y}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}] \mathbf{U}(s)$$

$$\boxed{\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}}$$

For SISO systems ($\mathbf{D} = 0$ typically): $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$

The **characteristic polynomial** is $\det(s\mathbf{I} - \mathbf{A})$, whose roots are the **eigenvalues** of $\mathbf{A} =$ poles of the system.

Example: TF from State Space

Given: $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $\mathbf{D} = 0$

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}, \quad \det = s^2 + 3s + 2$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{s^2 + 3s + 2}$$

Exam Tips — State Space & STM

- **STM** = $e^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$ — Laplace method is fastest in exams.
- **Homogeneous solution:** $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$ — zero input, only initial state.
- **Non-homogeneous solution:** zero-input + zero-state (convolution integral).
- **TF from SS:** $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$ — set $\mathbf{x}(0) = \mathbf{0}$.
- **Poles** of the system = eigenvalues of $\mathbf{A}$ = roots of $\det(s\mathbf{I} - \mathbf{A}) = 0$.
- $\Phi(0) = \mathbf{I}$ always — a quick check to verify your STM computation.

6. Controllability

Controllability: A system is said to be **completely state controllable** if it is possible to transfer the system from *any* initial state $\mathbf{x}(t_0)$ to *any* desired final state $\mathbf{x}(t_f)$ in a finite time $t_f - t_0$ using an unconstrained control input $\mathbf{u}(t)$.

In simple terms: *Can the input reach and control every state variable?*

6.1 Kalman's Controllability Test

Controllability Matrix

The system $(\mathbf{A}, \mathbf{B})$ is **completely controllable** if and only if the **controllability matrix** $\mathbf{Q}_c$ has full rank n :

$$\mathbf{Q}_c = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}]$$

$$\text{rank}(\mathbf{Q}_c) = n \Leftrightarrow \det(\mathbf{Q}_c) \neq 0 \Leftrightarrow \text{System is controllable}$$

$\mathbf{Q}_c$ is an $n \times nr$ matrix (for SISO: $n \times n$ square matrix).

Example: Check Controllability

Given: $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $n = 2$

$$\mathbf{AB} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\mathbf{Q}_c = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix}$$

$$\det(\mathbf{Q}_c) = (0)(-3) - (1)(1) = -1 \neq 0$$

Result: $\text{rank}(\mathbf{Q}_c) = 2 = n \Rightarrow$ System is **completely controllable**.

6.2 Physical Interpretation

- If a system is **not controllable**, some state variables are *decoupled* from the input — no matter what $\mathbf{u}(t)$ is applied, those states cannot be driven to a desired value.
- **Pole placement** (assigning closed-loop poles arbitrarily) is possible *only if* the system is completely controllable.
- An uncontrollable mode corresponds to a pole of the system that cannot be moved by state feedback.

7. Observability

Observability: A system is said to be **completely observable** if every state variable $x_i(t_0)$ can be determined from the output $y(t)$ over a finite time interval $[t_0, t_f]$, given the input $u(t)$.

In simple terms: *Can we reconstruct every state variable by observing the output?*

7.1 Kalman's Observability Test

Observability Matrix

The system (A, C) is **completely observable** if and only if the **observability matrix** Q_o has full rank n :

$$Q_o = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

$$\text{rank}(Q_o) = n \Leftrightarrow \det(Q_o) \neq 0 \Leftrightarrow \text{System is observable}$$

Q_o is an $mn \times n$ matrix (for SISO: $n \times n$ square matrix).

Example: Check Observability

Given: $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $n = 2$

$$CA = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$Q_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det(Q_o) = 1 \neq 0$$

Result: $\text{rank}(Q_o) = 2 = n \Rightarrow$ System is **completely observable**.

7.2 Physical Interpretation

- If a system is **not observable**, some state variables are *hidden* from the output — they cannot be estimated from measurements.
- **State observer (Luenberger observer)** design is possible *only if* the system is completely observable.
- An unobservable mode corresponds to a pole that does not appear in the transfer function (pole-zero cancellation).

7.3 Controllability vs Observability — Duality

Duality Principle (Kalman)

The system $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ is **controllable** if and only if the **dual system** $(\mathbf{A}^T, \mathbf{C}^T, \mathbf{B}^T)$ is **observable**, and vice versa.

This means: $\mathbf{Q}_c(\mathbf{A}, \mathbf{B}) = \mathbf{Q}_o^T(\mathbf{A}^T, \mathbf{C}^T)$

Property	Controllability	Observability
Definition	Transfer any state to any other	Determine any state from output
Test matrix	$\mathbf{Q}_c = [\mathbf{B} \ \mathbf{A}\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$	$\mathbf{Q}_o = [\mathbf{C}^T \ (\mathbf{C}\mathbf{A})^T \ \dots]^T$
Condition	$\text{rank}(\mathbf{Q}_c) = n$	$\text{rank}(\mathbf{Q}_o) = n$
Matrices involved	$\mathbf{A}$ and $\mathbf{B}$	$\mathbf{A}$ and $\mathbf{C}$
Application	Pole placement, state feedback	State observer (Luenberger)
Dual of	Observability of $(\mathbf{A}^T, \mathbf{C}^T)$	Controllability of $(\mathbf{A}^T, \mathbf{C}^T)$

Exam Tips — Controllability & Observability

- **Controllability matrix:** $\mathbf{Q}_c = [\mathbf{B} \ \mathbf{A}\mathbf{B} \ \mathbf{A}^2\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$ — involves $\mathbf{A}$ and $\mathbf{B}$.
- **Observability matrix:** $\mathbf{Q}_o = [\mathbf{C}^T \ (\mathbf{C}\mathbf{A})^T \ \dots \ (\mathbf{C}\mathbf{A}^{n-1})^T]^T$ — involves $\mathbf{A}$ and $\mathbf{C}$.
- Both tests: compute the matrix, find its determinant. Non-zero $\Rightarrow$ full rank $\Rightarrow$ controllable/observable.
- **Pole-zero cancellation** in TF $\Rightarrow$ system is either uncontrollable or unobservable (or both).
- **Duality:** $(\mathbf{A}, \mathbf{B})$ controllable $\Leftrightarrow (\mathbf{A}^T, \mathbf{B}^T)$ observable.
- For a system in **phase variable form**: always controllable; observable depends on $\mathbf{C}$.

Quick Revision — Module 5

State Variable Analysis | EC601

Key Formulas at a Glance

State equation	$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$
Output equation	$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$
STM definition	$\Phi(t) = e^{\mathbf{A}t}$
STM via Laplace	$\Phi(t) = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$
STM series	$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2 t^2}{2!} + \dots$
Homogeneous solution	$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$
Non-homogeneous solution	$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau$
Transfer function from SS	$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$
Characteristic polynomial	$\det(s\mathbf{I} - \mathbf{A}) = 0$
Controllability matrix	$\mathbf{Q}_c = [\mathbf{B} \ \mathbf{A}\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$
Controllable condition	$\text{rank}(\mathbf{Q}_c) = n$
Observability matrix	$\mathbf{Q}_o = [\mathbf{C}^T \ (\mathbf{C}\mathbf{A})^T \ \dots \ (\mathbf{C}\mathbf{A}^{n-1})^T]^T$
Observable condition	$\text{rank}(\mathbf{Q}_o) = n$
STM property	$\Phi(0) = \mathbf{I}; \Phi^{-1}(t) = \Phi(-t)$

One-Line Definitions

- **State:** Minimum set of variables that, with input, completely determines future system behaviour.
- **State Variable:** Each member of the minimum set; n variables for an n -th order system.
- **State Vector $\mathbf{x}$:** Column vector of all n state variables.
- **State Space:** n -dimensional space with state variables as axes.
- **A matrix:** System/plant matrix ($n \times n$); its eigenvalues are the system poles.
- **B matrix:** Input matrix ($n \times r$); couples input to state derivatives.
- **C matrix:** Output matrix ($m \times n$); selects which states appear in output.
- **D matrix:** Direct transmission matrix; usually zero for physical systems.
- **STM $\Phi(t)$:** Matrix exponential $e^{\mathbf{A}t}$; maps initial state to state at time t .
- **Homogeneous solution:** Zero-input response; $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$.

- **Non-homogeneous solution:** Zero-input + zero-state (convolution with input).
- **Controllability:** Ability to drive any state to any desired value using input.
- **Observability:** Ability to determine any state from output measurements.
- **Duality:** $(\mathbf{A}, \mathbf{B})$ controllable $\Leftrightarrow (\mathbf{A}^T, \mathbf{C}^T)$ observable.

Frequently Asked / Viva Questions

Q1. What is a state variable?

The minimum set of variables that, together with the input, completely determines the future behaviour of the system. An n -th order system has exactly n state variables.

Q2. What is the State Transition Matrix?

$\Phi(t) = e^{\mathbf{A}t}$. It maps the initial state to the state at time t in the absence of input: $\mathbf{x}(t) = \Phi(t)\mathbf{x}(0)$.

Q3. How is the STM computed using Laplace transform?

$\Phi(t) = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$. Form $(s\mathbf{I} - \mathbf{A})$, invert it, then take the inverse Laplace of each element.

Q4. What is the solution of the homogeneous state equation?

$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$. This is the zero-input (free) response depending only on the initial state.

Q5. What is the solution of the non-homogeneous state equation?

$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau$. Sum of zero-input and zero-state (convolution) responses.

Q6. How is the transfer function obtained from state space?

Set $\mathbf{x}(0) = \mathbf{0}$ and take Laplace transform: $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$.

Q7. What are the poles of a state space system?

The eigenvalues of the $\mathbf{A}$ matrix, i.e., roots of $\det(s\mathbf{I} - \mathbf{A}) = 0$ (characteristic equation).

Q8. Define controllability and state its test.

A system is controllable if any state can be driven to any desired state in finite time. Test: $\text{rank}(\mathbf{Q}_c) = n$ where $\mathbf{Q}_c = [\mathbf{B} \ \mathbf{A}\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$.

Q9. Define observability and state its test.

A system is observable if every state can be determined from the output in finite time. Test: $\text{rank}(\mathbf{Q}_o) = n$ where $\mathbf{Q}_o = [\mathbf{C}^T \ (\mathbf{C}\mathbf{A})^T \ \dots]^T$.

Q10. What is the duality principle?

$(\mathbf{A}, \mathbf{B})$ is controllable if and only if $(\mathbf{A}^T, \mathbf{B}^T)$ is observable. Controllability and observability are dual concepts.

Frequently Asked / Viva Questions (contd.)

Q11. What does pole-zero cancellation in TF indicate?

It indicates the system is either uncontrollable or unobservable (or both). The cancelled pole is a mode that is decoupled from either the input or the output.

Q12. What are the properties of the STM?

- $\Phi(0) = \mathbf{I}$ (identity at $t = 0$)
- $\Phi^{-1}(t) = \Phi(-t)$
- $\Phi(t_1 + t_2) = \Phi(t_1) \Phi(t_2)$
- $\dot{\Phi}(t) = \mathbf{A} \Phi(t)$

State Space Matrices — Quick Reference

Matrix	Size	Name	Role
A	$n \times n$	System matrix	Governs state dynamics; eigenvalues = poles
B	$n \times r$	Input matrix	Couples input to state derivatives
C	$m \times n$	Output matrix	Selects states visible in output
D	$m \times r$	Feedthrough matrix	Direct input-to-output path (usually $\mathbf{0}$)
$\mathbf{Q}_c$	$n \times nr$	Controllability matrix	Full rank $\Rightarrow$ controllable
$\mathbf{Q}_o$	$mn \times n$	Observability matrix	Full rank $\Rightarrow$ observable

Module 6

Nonlinear Control Systems — Basic Concepts

1. Nonlinear Control Systems — Basic Concepts

Real-world control systems are inherently nonlinear. Linear analysis (transfer functions, Bode plots, root locus) applies only when the system is linearised around an operating point. When nonlinearities are significant, dedicated nonlinear analysis methods are required.

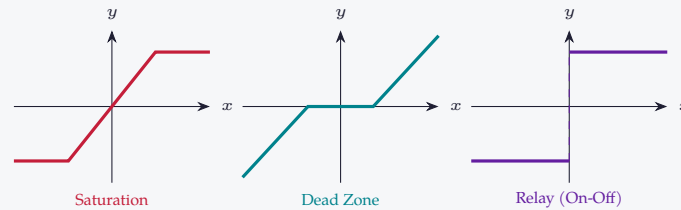
Nonlinear System: A system that does not satisfy the principle of superposition (additivity + homogeneity). Its behaviour depends on the *magnitude* and *nature* of the input, not just its frequency.

Key difference from linear systems: Frequency of output may differ from input; multiple equilibrium points possible; limit cycles can exist; stability depends on initial conditions.

1.1 Common Nonlinearities in Control Systems

Nonlinearity	Description	Example
Saturation	Output limited to a maximum value regardless of input magnitude	Amplifier, actuator
Dead Zone	No output for small inputs below a threshold	Gear backlash, friction
Backlash	Hysteresis in mechanical coupling; output lags input reversal	Gear trains
Relay (On-Off)	Output switches between two fixed values	Thermostat, bang-bang control
Coulomb Friction	Constant opposing force independent of velocity	Sliding surfaces
Hysteresis	Output depends on history of input	Magnetic materials

Common Nonlinearity Characteristics



1.2 Analysis Methods for Nonlinear Systems

- **Phase plane analysis:** Graphical method for 2nd order systems; plots $\dot{x}$ vs x .
- **Describing function method:** Frequency-domain approximation for nonlinear elements.
- **Lyapunov's method:** Energy-based stability analysis without solving the equations.
- **Linearisation:** Valid only near an operating point; uses Taylor series expansion.

2. Describing Function Method

Describing Function (DF): An approximate frequency-domain representation of a nonlinear element. It is defined as the ratio of the *fundamental harmonic component* of the output to the sinusoidal input, assuming the input is a pure sinusoid $x(t) = A \sin(\omega t)$.

$$N(A) = \frac{B_1 + jA_1}{A}$$

where A_1 and B_1 are the cosine and sine coefficients of the fundamental harmonic of the output.

2.1 Assumptions of the Describing Function Method

1. The nonlinear element is **single-valued** (no memory/hysteresis) or has known hysteresis.
2. The input to the nonlinear element is a **pure sinusoid**: $x(t) = A \sin(\omega t)$.
3. The linear part of the system has **sufficient low-pass filtering** so that higher harmonics of the output are attenuated — only the fundamental harmonic reaches the nonlinear element.
4. The nonlinear element is **odd-symmetric**: $f(-x) = -f(x)$ (no even harmonics).

2.2 Describing Functions of Common Nonlinearities

Nonlinearity	Describing Function $N(A)$	Notes
Ideal Relay	$\frac{4M}{\pi A}$	M = relay output level; real, decreases with A
Saturation	$\frac{2K}{\pi} \left[\sin^{-1}\left(\frac{D}{A}\right) + \frac{D}{A} \sqrt{1 - \left(\frac{D}{A}\right)^2} \right]$	K = slope, D = saturation level; valid for $A > D$
Dead Zone	$K \left[1 - \frac{2}{\pi} \sin^{-1}\left(\frac{\Delta}{A}\right) - \frac{2\Delta}{\pi A} \sqrt{1 - \left(\frac{\Delta}{A}\right)^2} \right]$ valid for $A > \Delta$	(Δ = dead zone half-width;
Backlash	Complex (has imaginary part)	Phase lag introduced; $N(A)$ is complex

2.3 Limit Cycle Prediction Using Describing Function

The describing function method is primarily used to predict **limit cycles** — sustained oscillations in nonlinear systems.

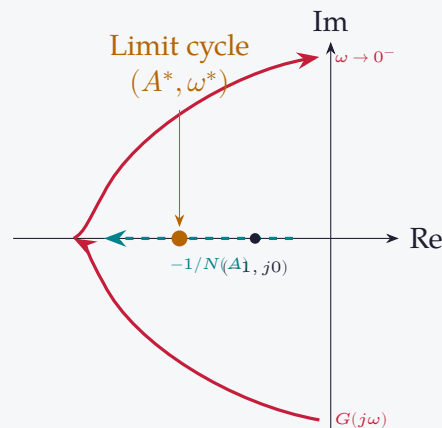
Limit Cycle Condition

For a nonlinear system with nonlinear element $N(A)$ in the forward path and linear element $G(j\omega)$ in the loop, a limit cycle exists when:

$$G(j\omega) \cdot N(A) = -1 \quad \Rightarrow \quad G(j\omega) = -\frac{1}{N(A)}$$

Graphical method: Plot $G(j\omega)$ (Nyquist plot) and $-1/N(A)$ (negative inverse DF locus) on the same complex plane. Their **intersection** gives the limit cycle amplitude A^* and frequency ω^* .

Limit Cycle Prediction — Graphical Method



Exam Tips — Describing Function

- DF is an **approximation** — valid only when higher harmonics are filtered by the linear part.
- DF of ideal relay: $N(A) = 4M/(\pi A)$ — purely real, decreases as A increases.
- **Limit cycle** exists at intersection of $G(j\omega)$ and $-1/N(A)$ loci.
- If $-1/N(A)$ locus does **not intersect** $G(j\omega)$: no limit cycle predicted.
- **Stability of limit cycle:** If increasing A moves $-1/N(A)$ outside $G(j\omega)$, the limit cycle is stable.
- DF method cannot detect **subharmonic oscillations** or **chaotic behaviour**.

3. Introduction to Optimal Control

Optimal Control: A control strategy that minimises (or maximises) a specified **performance index** (cost function) J while satisfying the system dynamics and constraints. It provides the *best possible* control in a mathematically defined sense.

Performance Index (Cost Function): A scalar measure of system performance. Common form:

$$J = \int_0^{t_f} L(\mathbf{x}(t), \mathbf{u}(t), t) dt + \phi(\mathbf{x}(t_f))$$

where L is the running cost and ϕ is the terminal cost.

3.1 Quadratic Performance Index (LQR)

The most widely used performance index for linear systems is the **quadratic** form:

Quadratic Performance Index

$$J = \int_0^{\infty} [\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) + \mathbf{u}^T(t) \mathbf{R} \mathbf{u}(t)] dt$$

where:

- $\mathbf{Q}$ ($n \times n$, positive semi-definite) — penalises state deviation from zero
- $\mathbf{R}$ ($r \times r$, positive definite) — penalises control effort
- Large $\mathbf{Q}$: fast state regulation (aggressive control)
- Large $\mathbf{R}$: small control effort (gentle control)

The optimal control law is: $\mathbf{u}^*(t) = -\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}\mathbf{x}(t) = -\mathbf{K}\mathbf{x}(t)$

where $\mathbf{P}$ satisfies the **Algebraic Riccati Equation (ARE)**:

$$\mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} + \mathbf{Q} = \mathbf{0}$$

3.2 Regulator Problem

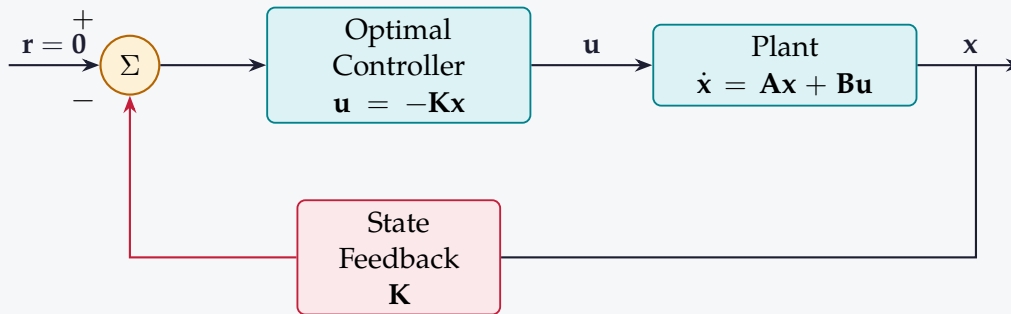
Regulator Problem: Drive the state $\mathbf{x}(t)$ to **zero** (or maintain it at zero) in the presence of disturbances, starting from a non-zero initial condition $\mathbf{x}(0) \neq \mathbf{0}$.

Goal: $\mathbf{x}(t) \rightarrow \mathbf{0}$ as $t \rightarrow \infty$, while minimising the cost J .

Reference: $\mathbf{r}(t) = \mathbf{0}$ (zero reference). The system must reject disturbances and return to equilibrium.

Example: Stabilising an inverted pendulum at the upright position.

Regulator Problem Block Diagram



3.3 Output Regulator Problem

Output Regulator Problem: A variant of the regulator problem where the **output** $\mathbf{y}(t)$ (not the full state) is driven to zero. The performance index penalises the output rather than the full state:

$$J = \int_0^{\infty} [\mathbf{y}^T \mathbf{Q}_y \mathbf{y} + \mathbf{u}^T \mathbf{R} \mathbf{u}] dt = \int_0^{\infty} [\mathbf{x}^T \mathbf{C}^T \mathbf{Q}_y \mathbf{C} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}] dt$$

This is equivalent to the state regulator with $\mathbf{Q} = \mathbf{C}^T \mathbf{Q}_y \mathbf{C}$.

Use case: When only the output (not all states) needs to be regulated, and full state measurement is unavailable.

3.4 Tracking Problem

Tracking Problem: The output $\mathbf{y}(t)$ must **follow** (track) a desired reference trajectory $\mathbf{r}(t) \neq \mathbf{0}$. The error $\mathbf{e}(t) = \mathbf{r}(t) - \mathbf{y}(t)$ must be minimised.

Performance index:

$$J = \int_0^{t_f} [\mathbf{e}^T(t) \mathbf{Q} \mathbf{e}(t) + \mathbf{u}^T(t) \mathbf{R} \mathbf{u}(t)] dt$$

Example: Robot arm tracking a desired trajectory; missile tracking a moving target; CNC machine following a path.

Problem	Reference	Goal	Example
Regulator	$\mathbf{r} = \mathbf{0}$	$\mathbf{x}(t) \rightarrow \mathbf{0}$	Stabilise inverted pendulum
Output Regulator	$\mathbf{r} = \mathbf{0}$	$\mathbf{y}(t) \rightarrow \mathbf{0}$	Regulate output only
Tracking	$\mathbf{r}(t) \neq \mathbf{0}$	$\mathbf{y}(t) \rightarrow \mathbf{r}(t)$	Robot arm trajectory

Exam Tips — Optimal Control

- **LQR** = Linear Quadratic Regulator; optimal for linear systems with quadratic cost.
- **Q large** $\Rightarrow$ fast state regulation; **R large** $\Rightarrow$ small control effort.
- Optimal gain: $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}$ where $\mathbf{P}$ solves the Algebraic Riccati Equation.
- **Regulator**: zero reference, drive state to zero. **Tracking**: non-zero reference, follow trajectory.
- **Output regulator**: only output penalised; equivalent to state regulator with $\mathbf{Q} = \mathbf{C}^T\mathbf{Q}_y\mathbf{C}$.
- Optimal control requires the system to be **controllable** (otherwise J cannot be minimised).

Quick Revision — Module 6

Nonlinear Systems & Optimal Control | EC601

Key Formulas at a Glance

DF of ideal relay	$N(A) = 4M/(\pi A)$
Limit cycle condition	$G(j\omega) = -1/N(A)$
Quadratic cost function	$J = \int_0^\infty (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt$
Optimal control law (LQR)	$\mathbf{u}^* = -\mathbf{K} \mathbf{x} = -\mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} \mathbf{x}$
Algebraic Riccati Eq.	$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} + \mathbf{Q} = 0$
Output regulator Q	$\mathbf{Q} = \mathbf{C}^T \mathbf{Q}_y \mathbf{C}$
Tracking error	$\mathbf{e}(t) = \mathbf{r}(t) - \mathbf{y}(t)$

One-Line Definitions

- **Nonlinear system:** Does not satisfy superposition; output depends on input magnitude.
- **Saturation:** Output limited to a maximum value regardless of input.
- **Dead zone:** No output for inputs below a threshold.
- **Limit cycle:** Sustained periodic oscillation in a nonlinear system.
- **Describing Function:** Ratio of fundamental harmonic output to sinusoidal input; $N(A)$.
- **Optimal control:** Minimises a performance index J while satisfying system dynamics.
- **LQR:** Linear Quadratic Regulator; optimal state feedback for linear systems.
- **Regulator problem:** Drive state to zero from non-zero initial condition.
- **Output regulator:** Drive output (not full state) to zero.
- **Tracking problem:** Output must follow a non-zero reference trajectory.
- **Riccati equation:** Matrix equation whose solution $\mathbf{P}$ gives the optimal gain $\mathbf{K}$.

Frequently Asked / Viva Questions

Q1. What is a nonlinear system?

A system that does not satisfy superposition (additivity + homogeneity). Its behaviour depends on input magnitude and initial conditions.

Q2. Name four common nonlinearities in control systems.

Frequently Asked / Viva Questions (contd.)

Saturation, dead zone, backlash, relay (on-off), Coulomb friction, hysteresis.

Q3. What is the Describing Function?

An approximate frequency-domain representation of a nonlinear element. $N(A)$ = ratio of fundamental harmonic output to sinusoidal input amplitude A .

Q4. What is the DF of an ideal relay?

$N(A) = 4M/(\pi A)$, where M is the relay output level. It is purely real and decreases as A increases.

Q5. How is a limit cycle predicted using the DF method?

Plot $G(j\omega)$ (Nyquist) and $-1/N(A)$ on the same complex plane. Their intersection gives the limit cycle amplitude A^* and frequency ω^* .

Q6. What are the assumptions of the DF method?

Sinusoidal input to nonlinear element; linear part has sufficient low-pass filtering; nonlinearity is odd-symmetric; single-valued (or known hysteresis).

Q7. What is the optimal control problem?

Find the control $\mathbf{u}(t)$ that minimises a performance index J subject to the system dynamics $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$.

Q8. What is the LQR cost function?

$J = \int_0^\infty (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt$. $\mathbf{Q}$ penalises state deviation; $\mathbf{R}$ penalises control effort.

Q9. What is the regulator problem?

Drive the state $\mathbf{x}(t) \rightarrow \mathbf{0}$ from a non-zero initial condition, with zero reference input, while minimising J .

Q10. What is the difference between regulator and tracking?

Regulator: reference is zero, goal is to drive state/output to zero. Tracking: reference is non-zero, goal is to follow the reference trajectory.

Q11. What is the Algebraic Riccati Equation?

$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} + \mathbf{Q} = \mathbf{0}$. Its solution $\mathbf{P}$ gives the optimal gain $\mathbf{K} = \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P}$.

Module 7

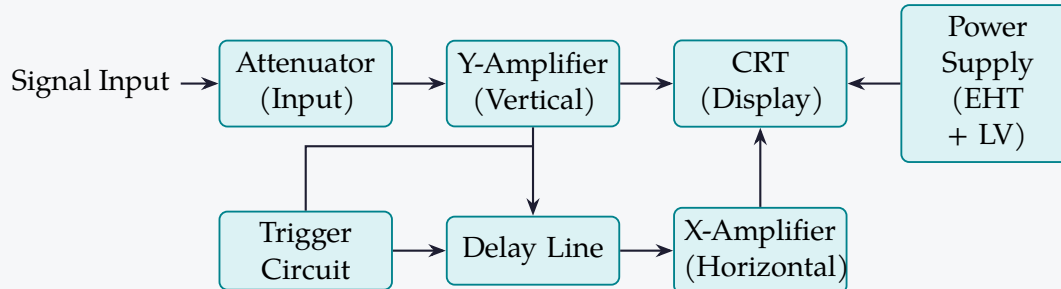
Cathode Ray Oscilloscope (CRO)

1. Cathode Ray Oscilloscope (CRO)

CRO (Cathode Ray Oscilloscope): An electronic test instrument that displays electrical signals as waveforms on a phosphor screen. It plots voltage (Y-axis) vs time (X-axis) in real time, allowing measurement of amplitude, frequency, phase, rise time, and waveform shape.

1.1 Block Diagram of CRO

CRO Block Diagram



1.2 Main Blocks and Functions

Block	Function
Attenuator	Reduces input signal to a safe level for the amplifier. Calibrated in V/div.
Y-Amplifier	Amplifies the vertical (signal) input to drive the vertical deflection plates of CRT.
Delay Line	Delays the signal slightly so the sweep starts before the signal arrives — ensures complete waveform display.
Trigger Circuit	Synchronises the sweep with the input signal to produce a stable display.
Time Base (Sweep Gen.)	Generates a sawtooth waveform for horizontal deflection. Calibrated in time/div.
X-Amplifier	Amplifies the time base signal to drive the horizontal deflection plates.
CRT	Cathode Ray Tube — electron gun, deflection plates, phosphor screen. Displays the waveform.
Power Supply	Provides EHT (~1–10 kV) for CRT and low voltages for amplifiers.

1.3 Measurements with CRO

Measurement	Method
Voltage (peak)	Count vertical divisions $\times$ V/div setting. $V_{pk} = n_{div} \times V/div$
Time period T	Count horizontal divisions for one cycle $\times$ time/div. $T = n_{div} \times \text{time/div}$
Frequency	$f = 1/T$
Phase difference	Use dual-trace mode; measure time shift Δt . $\phi = (\Delta t/T) \times 360^\circ$
Rise time	Time for signal to go from 10% to 90% of final value.
DC voltage	Set coupling to DC; measure vertical displacement from zero.

1.4 Lissajous Figures (Phase & Frequency Measurement)

Lissajous figures are patterns displayed on a CRO screen when two sinusoidal signals of different or same frequencies are applied simultaneously to the horizontal (X) and vertical (Y) deflection plates. They are used to measure **phase difference** and **frequency ratio**.

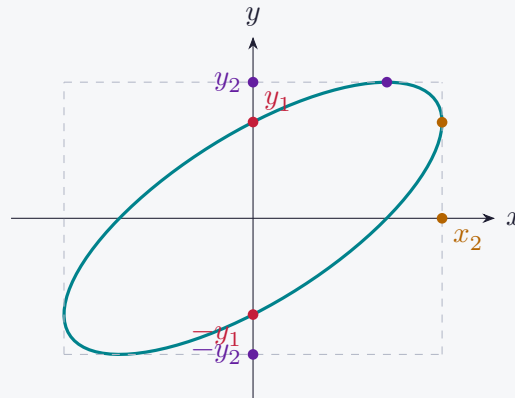
Phase Difference Measurement

When two sine waves of equal frequency ($f_x = f_y$) but different phase are applied, the pattern is an **ellipse**. The phase difference θ is calculated from the intercepts:

$$\sin \theta = \frac{y_1}{y_2} \Rightarrow \theta = \sin^{-1}\left(\frac{y_1}{y_2}\right) \quad \text{or} \quad 180^\circ - \sin^{-1}\left(\frac{y_1}{y_2}\right)$$

where y_1 is the y-intercept (height at $x = 0$) and y_2 is the maximum vertical deflection.

Lissajous Phase Ellipse



Frequency Measurement

If the frequencies are unequal, the frequency ratio is determined by drawing a horizontal and a vertical line through the pattern and counting the number of tangency points:

$$\frac{f_y}{f_x} = \frac{\text{Number of Horizontal Tangencies } (N_h)}{\text{Number of Vertical Tangencies } (N_v)}$$

1.5 Special Purpose Oscilloscopes: Dual-Trace vs. Dual-Beam & DSO

Modern electronic testing requires displaying multiple signals simultaneously and storing waveforms for analysis.

1.5.1 Dual-Trace vs. Dual-Beam Oscilloscopes

Parameter	Dual-Trace CRO	Dual-Beam CRO
Electron Gun	Single electron gun.	Two separate electron guns (or split beam with two deflection systems).
Deflection System	One set of vertical deflection plates.	Two independent sets of vertical deflection plates.
Working Method	Uses electronic switching (Chop or Alternate mode) to display two traces.	Displays two signals simultaneously by writing with two separate beams.
Modes	Chop: switches channels at high speed (low f signals). Alternate: swaps channel after each sweep (high f signals).	Simpler circuitry; no switching required.
Cost & Complexity	Moderate cost; standard design.	Highly complex; expensive CRT assembly.
Trace Alignment	Perfect alignment; single electron path.	Beam alignment can drift over time.

1.5.2 Digital Storage Oscilloscope (DSO)

A **Digital Storage Oscilloscope (DSO)** digitises the analog input signal and stores it in digital memory. It then reads this data from memory to reconstruct and display the waveform on the screen.

DSO Digitisation Block Diagram



- **Sample and Hold (S&H):** Holds the analog voltage constant long enough for the ADC to convert it.
- **ADC (Analog-to-Digital Converter):** Converts the held analog sample into a digital word (binary code).
- **Digital Memory (RAM):** Stores the binary samples sequentially.
- **DAC (Digital-to-Analog Converter):** Converts the stored data back to an analog signal.
- **Display:** Reconstructs and plots the signal.

Key Advantages of DSO over Analog CRO:

1. **Infinite Storage Time:** Waveforms are stored in memory and can be held indefinitely.
2. **Pre-trigger Viewing:** DSOs can display signals that occurred *before* the trigger point using a circular buffer.
3. **Signal Processing:** Direct microprocessor analysis (FFT, peak-to-peak, RMS, frequency).
4. **Easy Export:** Stored data can be transferred to a computer.

2. Wave Analyzer and Spectrum Analyzer

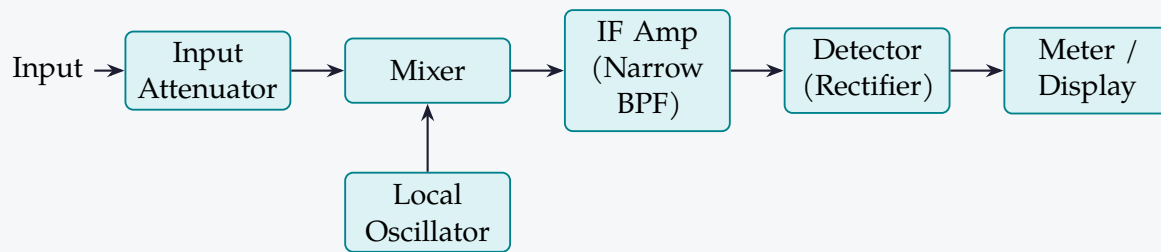
2.1 Wave Analyzer

Wave Analyzer: An instrument that measures the **amplitude** and **frequency** of individual harmonic components of a complex periodic waveform. It is essentially a *very narrow bandpass filter* that can be tuned to any frequency within its range.

Requirements:

- Very narrow, stable, and tunable bandpass filter
- High selectivity (ability to separate closely spaced harmonics)
- High sensitivity (detect small amplitude components)
- Calibrated frequency and amplitude readout

Wave Analyzer Block Diagram



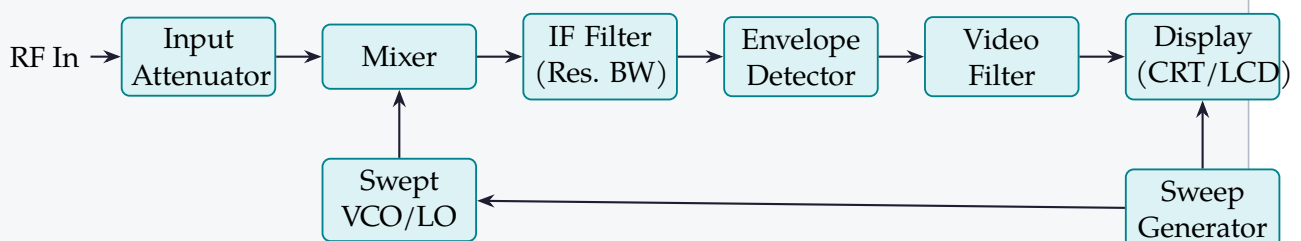
2.2 Spectrum Analyzer

Spectrum Analyzer: Displays the **frequency spectrum** of a signal — amplitude (or power) vs frequency. Unlike the wave analyzer (which measures one frequency at a time), the spectrum analyzer displays *all frequency components simultaneously* on a screen (frequency domain display).

Requirements:

- Wide frequency range with fine resolution
- High dynamic range (detect weak signals near strong ones)
- Fast sweep speed for real-time display
- Calibrated frequency (X-axis) and amplitude/power (Y-axis in dBm)

Spectrum Analyzer Block Diagram (Swept Superheterodyne)



Feature	Wave Analyzer	Spectrum Analyzer
Display	One frequency at a time	All frequencies simultaneously
Domain	Frequency (selective)	Frequency (full spectrum)
Speed	Slow (manual tuning)	Fast (swept or FFT)
Use	Harmonic distortion analysis	Wideband signal analysis

3. LVDT (Linear Variable Differential Transformer)

LVDT: A Linear Variable Differential Transformer is an electromechanical transducer that converts linear displacement into a proportional AC voltage. It is the most widely used inductive displacement sensor due to its high accuracy, infinite resolution, and frictionless operation.

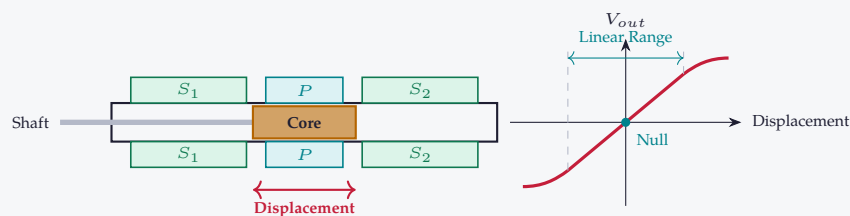
3.1 Construction and Working

Construction: One primary coil (centre) and two secondary coils (S1, S2) wound symmetrically on a cylindrical former. A movable ferromagnetic **core** (plunger) slides inside without contact.

Working:

1. AC excitation applied to primary coil induces EMF in both secondary coils.
2. Secondary coils connected in **series opposition**: $V_{out} = V_{S1} - V_{S2}$.
3. **Null position** (core centred): $V_{S1} = V_{S2}$, so $V_{out} = 0$.
4. **Core displaced right**: $V_{S1} > V_{S2}$, V_{out} is positive and in phase with primary.
5. **Core displaced left**: $V_{S2} > V_{S1}$, V_{out} is negative (180° phase shift).

LVDT Construction and Output Characteristic



Advantages	Disadvantages
Infinite resolution (frictionless)	Sensitive to stray magnetic fields
High accuracy and repeatability	Requires AC excitation and demodulator
No mechanical wear (contactless)	Limited to linear displacement
Rugged; works in harsh environments	Bulkier than resistive sensors

4. Servomotors

Servomotor: A motor specifically designed for use in **closed-loop control systems** (servo systems). It must have a linear torque-speed characteristic, low inertia, fast response, and be controllable by an error signal. Used for precise position, speed, or torque control.

4.1 DC Servomotor

DC Servomotor: A DC motor with a separately excited field or permanent magnet field, controlled by varying the **armature voltage**. The torque is proportional to armature current; speed is proportional to back-EMF.

DC Servomotor Equations

Electrical: $V_a = E_b + I_a R_a + L_a \frac{dI_a}{dt}$, $E_b = K_b \omega$

Mechanical: $T = K_T I_a = J \frac{d\omega}{dt} + B \omega$

Transfer Function (armature control, $L_a \approx 0$):

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_T/R_a}{Js + B + K_T K_b/R_a}$$

where K_T = torque constant, K_b = back-EMF constant, J = inertia, B = damping.

Feature	DC Servomotor
Control method	Armature voltage control or field control
Torque-speed	Linear (ideal for servo use)
Response	Fast; low inertia rotor
Maintenance	Brushes require periodic replacement
Applications	Robotics, CNC machines, antenna positioning

4.2 AC Servomotor

AC Servomotor: A two-phase induction motor used in servo systems. One phase (reference phase) is supplied with a fixed AC voltage; the other phase (control phase) receives a variable voltage proportional to the error signal. Speed and direction are controlled by the magnitude and phase of the control voltage.

AC Servomotor Key Features

- **High rotor resistance:** Ensures stable, linear torque-speed characteristic (negative slope throughout).
- **Low inertia:** Drag-cup or thin-disc rotor for fast response.
- **Control:** Vary amplitude of control phase voltage; direction by phase reversal ($\pm 90^\circ$).
- **Transfer function:** $G(s) = K_m / (s(\tau_m s + 1))$ where K_m = motor gain, τ_m = mechanical time constant.

Feature	DC Servomotor	AC Servomotor
Supply	DC	2-phase AC
Control	Armature voltage	Control phase voltage
Maintenance	Brushes needed	Brushless (robust)
Power range	High power	Low to medium power
Noise	More (brushes)	Less
Applications	Heavy-duty servo	Instruments, light servo

5. Tachogenerator (Tacho Generator)

Tachogenerator: A transducer that converts **rotational speed** into a proportional electrical voltage. Used as a **velocity feedback sensor** in closed-loop speed control systems.

DC Tacho: Output DC voltage proportional to speed: $V_{out} = K_t \omega$

AC Tacho: Output AC voltage whose amplitude is proportional to speed; frequency equals shaft rotation frequency.

Feature	DC Tachogenerator	AC Tachogenerator
Output	DC voltage $\propto$ speed	AC voltage amplitude $\propto$ speed
Direction sensing	Polarity reversal	Phase reversal
Ripple	Present (commutator)	None
Use	Speed feedback in DC drives	AC servo systems

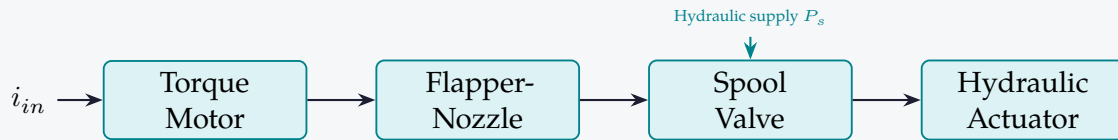
6. Hydraulic and Pneumatic Actuators

6.1 Electro-Hydraulic Valve

Electro-Hydraulic Valve (EHV): Converts an **electrical signal** (current or voltage) into a **hydraulic flow** or pressure. It controls the flow of hydraulic fluid to an actuator. Types: servo valve (proportional), solenoid valve (on-off).

Working: An electrical torque motor deflects a flapper between two nozzles, creating a differential pressure that positions a spool valve, which controls hydraulic flow to the actuator.

Electro-Hydraulic Servo Valve (Flapper-Nozzle Type)



6.2 Hydraulic Servomotor

Hydraulic Servomotor: A hydraulic actuator (linear cylinder or rotary motor) that converts **hydraulic power** (pressure $\times$ flow) into **mechanical motion**. Controlled by an electro-hydraulic valve.

Advantages: Very high force/torque output; fast response; high power-to-weight ratio.

Disadvantages: Requires hydraulic power supply; risk of leakage; maintenance intensive.

Hydraulic Actuator Transfer Function

For a hydraulic cylinder with spool valve:

$$\frac{X(s)}{I(s)} = \frac{K_h}{s(Ms + B)}$$

where K_h = hydraulic gain, M = load mass, B = viscous damping, X = piston displacement, I = input current to valve.

6.3 Electro-Pneumatic Valve

Electro-Pneumatic Valve: Converts an **electrical signal** into a **pneumatic pressure or flow**. Used to control pneumatic actuators. Types: I/P converter (current-to-pressure), solenoid valve.

I/P Converter: Converts 4–20 mA current signal to 3–15 psi (or 0.2–1.0 bar) pneumatic output. Standard interface between electronic controllers and pneumatic actuators in process industries.

6.4 Pneumatic Actuator

Pneumatic Actuator: Converts **compressed air pressure** into **mechanical motion** (linear or rotary). Controlled by an electro-pneumatic valve or I/P converter.

Types:

- **Spring-return diaphragm actuator:** Air pressure acts on a diaphragm against a spring. Used for control valves. Fail-safe (spring returns to safe position on air failure).
- **Double-acting cylinder:** Air applied alternately to both sides of a piston for bidirectional motion.

- **Rotary vane actuator:** Converts air pressure to rotary motion (limited angle).

Feature	Hydraulic	Pneumatic	Electric
Power density	Very high	Medium	Medium
Speed	Fast	Very fast	Fast
Precision	High	Moderate	Very high
Maintenance	High (leaks)	Low	Low
Compressibility	Incompressible	Compressible	N/A
Typical use	Heavy machinery	Process valves	Robotics, CNC

Quick Revision — Module 7

Instrumentation & Actuators | EC601

Key Formulas at a Glance

CRO voltage measurement	$V_{pk} = n_{div} \times V/div$
CRO time period	$T = n_{div} \times \text{time}/div$
CRO frequency	$f = 1/T$
CRO phase difference	$\phi = (\Delta t/T) \times 360^\circ$
Lissajous phase angle	$\sin \theta = y_1/y_2$
Lissajous frequency ratio	$f_y/f_x = N_h/N_v$
LVDT output	$V_{out} = V_{S1} - V_{S2}$; zero at null position
DC tachometer output	$V_{out} = K_t \omega$
DC servomotor TF	$\Omega(s)/V_a(s) = (K_T/R_a)/(Js + B + K_T K_b/R_a)$
AC servomotor TF	$G(s) = K_m/[s(\tau_m s + 1)]$
Hydraulic actuator TF	$X(s)/I(s) = K_h/[s(Ms + B)]$
I/P converter range	4–20 mA $\rightarrow$ 3–15 psi (0.2–1.0 bar)

One-Line Definitions

- **CRO:** Displays voltage vs time waveforms; measures amplitude, frequency, phase, rise time.
- **Trigger circuit:** Synchronises CRO sweep with input signal for stable display.
- **Delay line:** Delays signal in CRO so sweep starts before signal arrives.
- **Lissajous Figure:** Pattern displayed on CRO by two orthogonal sinusoidal signals.
- **Chop Mode:** Switching between channels at high speed; used for low-frequency signals.
- **Alternate Mode:** Switching channels after each sweep; used for high-frequency signals.
- **DSO:** Digital Storage Oscilloscope; digitises, stores, and displays waveforms.
- **Pre-trigger Viewing:** DSO feature to view signal before trigger using circular buffer.
- **Wave analyzer:** Measures amplitude of individual harmonics; narrow tunable BPF.
- **Spectrum analyzer:** Displays full frequency spectrum (amplitude vs frequency) simultaneously.

- **LVDT:** Inductive displacement sensor; output voltage proportional to core displacement.
- **Null position (LVDT):** Core centred; $V_{S1} = V_{S2}$; $V_{out} = 0$.
- **DC servomotor:** DC motor for servo use; controlled by armature voltage; linear torque-speed.
- **AC servomotor:** 2-phase induction motor; high rotor resistance; low inertia; drag-cup rotor.
- **Tachogenerator:** Speed-to-voltage transducer; used as velocity feedback sensor.
- **Electro-hydraulic valve:** Converts electrical signal to hydraulic flow; flapper-nozzle type.
- **Hydraulic servomotor:** High-force actuator; converts hydraulic power to mechanical motion.
- **I/P converter:** Converts 4–20 mA to 3–15 psi; interfaces electronic controller to pneumatic actuator.
- **Pneumatic actuator:** Converts compressed air to mechanical motion; spring-return is fail-safe.

Frequently Asked / Viva Questions

Q1. What is the function of the trigger circuit in a CRO?

It synchronises the time base sweep with the input signal so that the waveform appears stationary (stable) on the screen.

Q2. Why is a delay line used in a CRO?

To delay the signal slightly so the sweep generator starts before the signal arrives, ensuring the leading edge of the waveform is displayed.

Q3. How do you measure frequency using a CRO?

Measure the time period T (horizontal divisions $\times$ time/div for one complete cycle), then $f = 1/T$.

Q4. How can phase difference and frequency ratio be measured using Lissajous figures?

Phase difference: for equal frequencies, $\theta = \sin^{-1}(y_1/y_2)$ from the ellipse intercepts. Frequency ratio: for unequal frequencies, $f_y/f_x = N_h/N_v$, where N_h and N_v are horizontal and vertical tangency counts.

Q5. What is the difference between Chop and Alternate modes in a Dual-Trace CRO?

Chop mode switches between channels rapidly during a single sweep (best for low frequencies). Alternate mode sweeps Channel A completely, then Channel B on the next sweep (best for high frequencies to prevent trace flicker).

Q6. What is the main advantage of a DSO over an analog CRO?

Frequently Asked / Viva Questions (contd.)

DSOs digitise and store signals in digital memory, offering infinite waveform storage, pre-trigger viewing (circular buffer recording), and built-in microprocessor-based signal processing.

Q7. How does a DSO achieve pre-trigger viewing?

It continuously writes ADC samples to a circular memory buffer. When the trigger occurs, it stops writing after a specific delay, retaining the samples that were stored prior to the trigger event.

Q8. What is the difference between a wave analyzer and a spectrum analyzer?

Wave analyzer measures one frequency component at a time (narrow tunable BPF). Spectrum analyzer displays all frequency components simultaneously on screen.

Q9. What is LVDT and how does it work?

Linear Variable Differential Transformer. AC excitation on primary induces EMF in two secondary coils connected in series opposition. Output voltage is proportional to core displacement; zero at null position.

Q10. What is the output of LVDT at null position?

Zero, because the two secondary voltages are equal and opposite ($V_{S1} = V_{S2}$, so $V_{out} = V_{S1} - V_{S2} = 0$).

Q11. What are the advantages of LVDT?

Infinite resolution (frictionless/contactless), high accuracy, no mechanical wear, rugged, works in harsh environments.

Q12. What is the difference between DC and AC servomotor?

DC: controlled by armature voltage, uses brushes, high power. AC: 2-phase induction, brushless, high rotor resistance for linear torque-speed, low inertia drag-cup rotor.

Q13. What is a tachogenerator?

A speed transducer that generates a voltage proportional to rotational speed ($V = K_t \omega$). Used as velocity feedback in closed-loop speed control.

Q14. What is an electro-hydraulic servo valve?

Converts electrical current to hydraulic flow using a torque motor, flapper-nozzle, and spool valve. Controls hydraulic actuator position/speed.

Q15. What is an I/P converter?

Current-to-pressure converter. Converts 4–20 mA electrical signal to 3–15 psi pneumatic signal. Standard interface between electronic controllers and pneumatic actuators.

Q16. Why is a spring-return pneumatic actuator called fail-safe?

On loss of air supply, the spring returns the actuator (and connected valve) to a predetermined safe position (fully open or fully closed).